

Undecidability of translational monotilings in fixed dimension

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Abstract

For a fixed integer d , the translational monotiling problem TILE_d asks whether a given finite set $F \subseteq \mathbb{Z}^d$ tiles the whole lattice with multiplicity one. We prove that, for some absolute constant d_* , TILE_d is undecidable for every fixed $d \geq d_*$. Our reduction from the domino problem for finite Wang tile sets builds on the functional-equation framework of Greenfeld–Tao. Its main new idea is to store each complete local state, including its input-dependent Wang symbol, in one element of a finite extension field whose degree is fixed, although its characteristic may grow with the Wang instance. A finite-field rack makes an arbitrary state function available, and one exact-cover row excludes a prescribed state by testing whether multiplication by the difference between the stored state and the forbidden state permutes the field. Thus the size of the alphabet and the number of equations may grow while the numbers of group generators remain uniformly bounded, allowing the output dimension d_* to be chosen independently of the Wang instance.

1 Introduction

Let F be a finite subset of an abelian group G . We say that F *tiles G by translations* if there is a set $A \subseteq G$ such that $F \oplus A = G$, in the sense that every element of G has a unique representation $f + a$ with $f \in F$ and $a \in A$.

For $d \geq 1$, define

$$\text{TILE}_d := \{F \subseteq \mathbb{Z}^d : |F| < \infty \text{ and } \exists A \subseteq \mathbb{Z}^d \quad F \oplus A = \mathbb{Z}^d\}.$$

To formulate the decision problem precisely, we fix a computable encoding of finite subsets of $\mathbb{Z}^d$ by listing their integer vectors in lexicographic order and encoding their coordinates with a fixed computable scheme. We regard TILE_d as the language consisting of the encodings of those finite sets that tile $\mathbb{Z}^d$.

To establish undecidability, we reduce from the Wang tiling problem, which we now describe. An *encoding of a Wang instance* is a finite, possibly empty, list of quadruples of finitely encoded color labels in north, east, south, west order. We computably remove repeated quadruples, retaining their first occurrences; this preserves tileability because a deleted occurrence may always be replaced by its retained copy. Henceforth $\mathcal{W}$ denotes the resulting set of distinct tiles, ordered by first occurrence. A tiling assigns one tile in $\mathcal{W}$ to each point of $\mathbb{Z}^2$ so that east–west and north–south labels agree across every adjacent pair. Equality of color labels is part of the finite input and is decidable. Figure 1 shows how the four edge colors are displayed on each Wang tile and how the horizontal and vertical matching conditions propagate through a finite patch.

In the 1960s, Wang [23] introduced the domino problem as a finite combinatorial model for logical decision procedures. The finite local matching rules above can constrain an infinite configuration. Wang observed that tileability would be decidable if every tileable set admitted a periodic tiling, and conjectured that this periodicity property always holds. The logical background was sharpened by Kahr, Moore, and Wang [14], who reduced the general decision problem to the $\forall\exists\forall$ case.

Berger [2] refuted Wang’s expectation by encoding Turing-machine computations into local rules requiring the colors on adjacent tile edges to agree. From a machine in a family with an undecidable halting problem, his construction produces a finite tileset that tiles the plane exactly when the machine does not halt; hence no algorithm can decide the domino problem. Robinson [20] later gave a different construction proving undecidability and exhibiting finite tilesets that tile the plane but admit no periodic tiling; Jeandel and Vanier [13] give a modern account of these reductions. After a long search for small examples,

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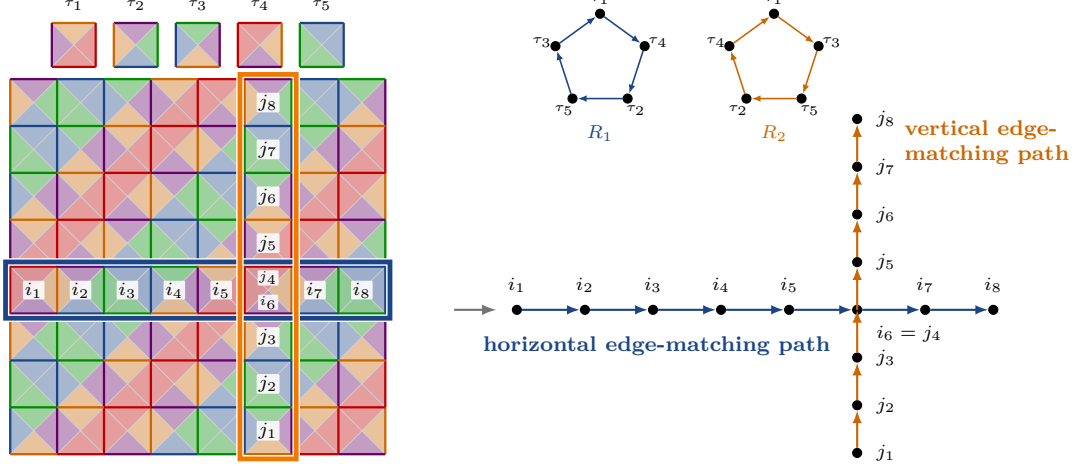


Figure 1: An 8×8 Wang patch and its matching paths. The highlighted row and column yield paths in the horizontal and vertical matching digraphs, meeting at the common state $i_6 = j_4$.

Jeandel and Rao [11] constructed an aperiodic set of 11 Wang tiles on four colors and proved minimality both in the number of tiles and in the number of colors.

For monotilings, the low-dimensional picture is different. Newman [18] proved that every tiling of $\mathbb{Z}$ by a finite tile is periodic. Coven and Meyerowitz [5] later developed cyclotomic criteria for such tilings, complete when the tile cardinality has at most two prime factors. Szegedy [22] gave algorithms for finite tiles of prime cardinality and of cardinality four, with extensions to finitely generated abelian groups. Bhattacharya [3] proved that every finite $F \subseteq \mathbb{Z}^2$ which tiles $\mathbb{Z}^2$ also admits a periodic tiling and, consequently, that TILE_2 is decidable. Greenfeld and Tao [7] subsequently obtained quantitative period bounds and stronger structural conclusions in dimension two. The periodic tiling conjecture, formulated in this setting by Lagarias and Wang [16], predicted that every finite tile which tiles $\mathbb{Z}^d$ also admits a periodic tiling. In a breakthrough result, Greenfeld and Tao [9] disproved this in sufficiently high dimension.

On the undecidability side, Ollinger [19] proved that translational tiling of the plane is undecidable even for sets of 11 polyominoes. Jeandel and Rolin [12] showed that the Wang tiling problem remains undecidable when the number of tiles exceeds the number of colors by at most 43. Greenfeld and Tao [8] developed a two-tile encoding framework and later proved [10] undecidability for translational monotilings when the finite quotient, periodic target, or lattice dimension is allowed to vary with the input. Recent work of Yang and Zhang [24] and Kim [15] establish fixed-dimensional multi-tile undecidability for, respectively, four tiles in $\mathbb{Z}^2$ and two connected tiles in $\mathbb{Z}^3$. We prove that a single input tile suffices while the lattice dimension is fixed in advance and the target is the whole lattice. This answers Question 6.3 in Greenfeld’s survey [6] and the third part of Question 34 posed by Coulbois, Gajardo, Guillon, and Lutfalla [4].

Theorem 1.1. *There is an absolute integer $d_* \geq 1$ and a total algorithm which, given an encoding of a finite Wang instance $\mathcal{W}$, outputs an encoding of a finite set $\hat{F}_{\mathcal{W}} \subseteq \mathbb{Z}^{d_*}$ such that $\mathcal{W}$ tiles $\mathbb{Z}^2$ if and only if $\hat{F}_{\mathcal{W}}$ tiles $\mathbb{Z}^{d_*}$. In particular, TILE_d is undecidable for every fixed integer $d \geq d_*$.*

The proof is effective in the usual computability-theoretic sense and makes no attempt to optimize d_* . The finite collection of structural fiberwise partition equations used below is chosen once and for all, independently of the Wang instance, and is included in the fixed reduction. Given the Wang instance, the remaining groups, equations, and tiles can all be constructed in finitely many steps.

The argument uses the weak-expressibility framework of Greenfeld–Tao [9] to build a fixed phase-pinned structural fiberwise partition system, and adds a finite-field construction that encodes all input-dependent local constraints while keeping the numbers of generators uniformly bounded. For the input-dependent part, this avoids binary expansion of the Wang alphabet, Boolean-circuit compilation, and a separate prime coordinate for every auxiliary variable. Input growth changes the field characteristic and the number of forbidden-state rows, but not the generator ranks. The final step applies a theorem of Meyerovitch, Sanadhya, and Solomon [17, Theorem 1.1] to convert the resulting tiling problem on a finitely generated

abelian group into an equivalent problem on the fixed lattice $\mathbb{Z}^{d*}$.

The proof proceeds in four stages.

1. We use a decorated q -adic Sudoku board to store an infinite sequence of vertices. At each q -adic scale the decoration reveals one vertex, and rescaling the exceptional rows reveals the next one. The line rule forces each consecutive pair to be a directed edge, so the successive scales trace a right-infinite directed walk.
2. We apply this construction in two independent directions. The first board enforces horizontal edge matching and the second enforces vertical edge matching. At each position their common decoration supplies one Wang tile that is seen by both boards, producing a tiling of the quadrant. Compactness then gives a tiling of the whole plane.
3. We encode the structural bits of the two boards and their common Wang decoration as a fixed number of coordinates over a finite field whose characteristic grows with the Wang instance. A fixed-degree extension represents each complete local state by a single field element. Fiberwise partition equations make an arbitrary field-valued state available, exclude every forbidden state by a permutation test, and identify its structural coordinates with the Sudoku fields. The field characteristic and the number of equations may grow, but the resulting fiberwise partition system uses uniformly bounded numbers of generators.
4. We regard a solution of the fiberwise partition system as its graph. The partition equations then become a finite family of tiles required to share this graph as a common tiling complement, and a rigid stacking construction combines the family into one tile. The uniform generator bound makes the resulting group a quotient of $\mathbb{Z}^{d*}$ for one fixed d_* . The theorem of Meyerovitch, Sanadhya, and Solomon then lifts this tile to an equivalent tile in $\mathbb{Z}^{d*}$.

The logical dependency chain, including the internal construction steps and the conversion to a single tile, is summarized in Figure 2.

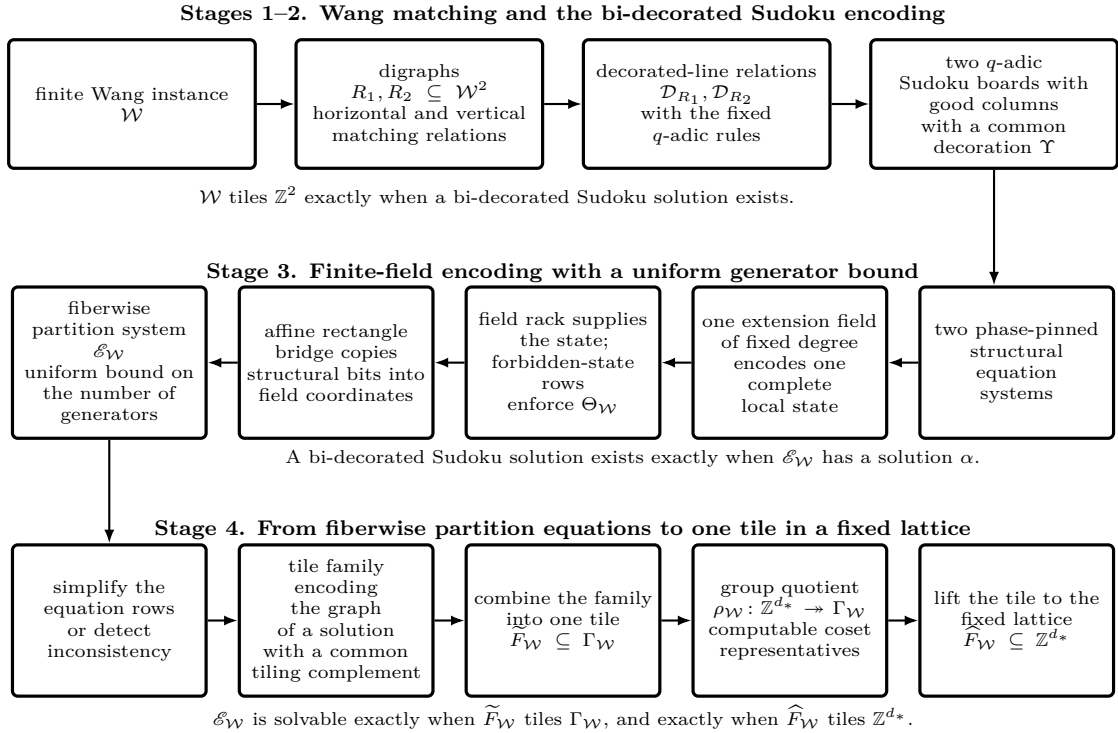


Figure 2: Roadmap of the reduction.

Section 2 introduces the fiberwise partition language. Section 3 constructs the decorated Sudoku encoding, and Sections 4 and 5 construct the bounded-generator fiberwise partition system. Section 6 converts that system to one tile and completes the reduction.

2 Fiberwise partition equations

All groups below are additive and abelian. Throughout, $\mathbb{N} = \{0, 1, 2, \dots\}$, and C_m denotes the cyclic additive group $(\mathbb{Z}/m\mathbb{Z}, +)$ for $m \geq 1$. Given additive abelian groups G and H , a map $\beta: G \rightarrow H$, and $h \in G$, define the discrete difference in direction h by $\partial_h \beta(x) := \beta(x + h) - \beta(x)$ for every $x \in G$.

Fix a finitely generated abelian group G and a finite abelian group H . Fix an integer $J \geq 1$, shifts $h_1, \dots, h_J \in G$, and finite blocks $E_1, \dots, E_J \subseteq H$. We say that a map $\alpha: G \rightarrow H$ satisfies the corresponding *fiberwise partition equation* if, for every $x \in G$,

$$\biguplus_{j=1}^J (\alpha(x + h_j) + E_j) = H.$$

A finite collection of such equations is a *fiberwise partition system*; a *solution* of the system is a map satisfying every equation in it. To explain the terminology, consider the projection $\pi: G \times H \rightarrow G$. Its fiber over x is $\{x\} \times H$, and the preceding identity says that the translates $\alpha(x + h_j) + E_j$ partition all of H . Thus the corresponding translates of the graph of α cover the entire fiber $\{x\} \times H$. Here “fiber” refers to the projection π , not to a level set of α .

For the algorithmic constructions below, we adopt the following encoding convention.

Definition 2.1. A *finite encoding of a fiberwise partition system* records the following information.

- (i) The domain G is given as a quotient $\mathbb{Z}^r/L$, with a finite integer generating matrix for the subgroup L .
- (ii) The codomain H , a finite abelian group, is given in invariant-factor form, and every quotient homomorphism appearing in the system is specified by explicit coordinate lists.
- (iii) Every shift h_j is given by an integer coordinate vector, and every block E_j by its complete finite list of elements of H .

Whenever we construct a fiberwise partition system algorithmically, its output consists of the presentations, coordinate vectors, and element lists specified above. Group operations and the specified quotient homomorphisms are computable from this encoding; equality, block membership, and equality of shifts are decidable by Smith normal form and finite enumeration.

A fixed finite family of groups, homomorphisms, shifts, and blocks in the format of Definition 2.1 can be included in the description of the reduction. After the variable finite input is supplied, the reduction performs only explicit calculations with integer presentations and finite element lists. The resulting map is therefore total and computable. The fixed family is chosen once and reused for every input.

Finitely many fiberwise partition systems for maps $\alpha_t: G \rightarrow H_t$ may be combined into one such system for $\alpha = (\alpha_t)_t: G \rightarrow \prod_t H_t$, with exactly the same solution tuples. Indeed, in a row belonging to coordinate t , replace every block $E_j \subseteq H_t$ by the cylinder $E_j \times \prod_{s \neq t} H_s$. These cylinders partition the product exactly when their H_t projections partition H_t .

We repeatedly use the following coordinate-cylinder identity.

Lemma 2.2. *Let G be a finitely generated abelian group, let H, L be finite abelian groups, let $\alpha: G \rightarrow H$, and let $\rho: H \twoheadrightarrow L$ be a quotient map. For any $h \in G$, the exact-cover identity*

$$(\alpha(x + h) + \ker \rho) \uplus (\alpha(x) + (H \setminus \ker \rho)) = H$$

holds for every $x \in G$ if and only if $\rho\alpha(x + h) = \rho\alpha(x)$ for every $x \in G$.

Proof. The second summand in the exact-cover identity is the complement of the coset $\alpha(x) + \ker \rho$. Exact coverage therefore forces the first coset to equal $\alpha(x) + \ker \rho$, which is precisely the quotient-period identity. The row imposes no condition within the cosets of $\ker \rho$. $\square$

If a map $\beta: G \rightarrow H$ satisfies $\beta(x + s) = \beta(x)$ for every $x \in G$ and $s \in S \subseteq G$, then β is invariant under the subgroup $\langle S \rangle$. Positive multiples follow by iterating the identities, negative multiples follow after replacing x by $x - s$, and arbitrary elements of $\langle S \rangle$ follow by composing these identities.

3 Decorated q -adic Sudoku

This section constructs a decorated q -adic Sudoku encoding of the matching relations in a Wang instance. We first make the finite decorated-line relation effective, then show that one board encodes a right-infinite directed walk, and finally synchronize two boards to encode the horizontal and vertical constraints of a Wang tiling.

The q -adic hierarchy used in our Sudoku encoding has its origin in the linear-sampling construction of Aanderaa and Lewis [1]. The formal ingredients needed here, including the q -adic Sudoku formulation and its normalization results, are taken from Greenfeld–Tao [9].

Following Greenfeld–Tao [9, Section 1.3], fix $s_0 = 10$ and set $q := 2^{s_0} = 1024$. Here $C_q \cong (\mathbb{Z}/q\mathbb{Z}, +)$. We identify C_q with the residue classes modulo q and use their natural multiplication when needed, and put $C_q^\circ := C_q \setminus \{0\}$. We will use the results from their Sections 7–9 listed below. Put $N := q^2$, write $[N] := \{1, \dots, N\}$, and set $\mathbb{B} := [N] \times \mathbb{Z}$. When convenient, we identify a function $\mu: [N] \rightarrow C_q^\circ$ with the tuple $(\mu(1), \dots, \mu(N)) \in (C_q^\circ)^N$. For a nonzero integer m , define $\nu_q(m) := \max\{s \in \mathbb{N}: q^s \mid m\}$, write $m = q^{\nu_q(m)}a$ with $q \nmid a$, and define $f_q(m) := a \pmod{q}$; set $f_q(0) := 1$. Following Greenfeld–Tao [9, Definition 7.3], let $\mathcal{S}_q[N] \subseteq (C_q^\circ)^N$ be the finite set of functions $\mu: [N] \rightarrow C_q^\circ$ for which there are integers a, b, c , with c odd, such that, for every $n \in [N]$,

$$\mu(n) = cf_q(an + b). \quad (3.1)$$

A *Sudoku solution* is a map $\Xi: \mathbb{B} \rightarrow C_q^\circ$ for which the function $n \mapsto \Xi(n, jn + i)$ lies in $\mathcal{S}_q[N]$ for every $i, j \in \mathbb{Z}$. It has *good columns* if, for each $n \in [N]$, there is a permutation $\sigma_n: C_q \rightarrow C_q$ such that $\Xi(n, m) = \sigma_n(m \bmod q)$ whenever $\sigma_n(m \bmod q) \neq 0$.

A Sudoku solution Ξ is in *normal form* if there is some $D \in C_q$ such that, for every $(n, m) \in \mathbb{B}$ with $q \nmid m$,

$$\Xi(n, m) = m + D \frac{q}{4} m(m - n) \pmod{q}. \quad (3.2)$$

Thus normal form determines the board on every row with $q \nmid m$ but places no restriction on the rows with $q \mid m$.

We use the following four results of Greenfeld–Tao.

- (i) Every $\mu \in \mathcal{S}_q[N]$ has a canonical nonzero affine map $\alpha_\mu: \mathbb{Z} \rightarrow C_q$. It also has an order $o_\mu \in \{-\infty, 0, \dots, s_0 - 1\}$ and a bad coset $\Gamma_\mu = \alpha_\mu^{-1}(0)$. One has $\mu(n) = \alpha_\mu(n)$ for every $n \in [N] \setminus \Gamma_\mu$, and, if $\Gamma_\mu \neq \emptyset$, then Γ_μ is a coset of $(q/2^{o_\mu})\mathbb{Z}$; see [9, Lemma 8.1 and Proposition 8.3].
- (ii) If Ξ is a Sudoku solution, then, for any integers a, b, c and any odd integer B , the maps $(n, m) \mapsto \Xi(n, am + bn + c)$ and $(n, m) \mapsto B\Xi(n, m)$ are also Sudoku solutions, where multiplication in the second map is taken modulo q ; see [9, Proposition 7.7].
- (iii) For every good-column Sudoku solution Ξ , there are integers A, C and an odd integer B such that the transformed board $\Xi'(n, m) := B\Xi(n, m + An + C)$ is a Sudoku solution in normal form. See [9, Lemma 9.1 and Propositions 9.4–9.6].
- (iv) If Ξ is in normal form, then there are integers A, C and an odd integer B for which the map Ξ^+ defined by

$$\Xi^+(n, m) := B\Xi(n, q(m + An + C)) \quad (3.3)$$

is again a Sudoku solution in normal form; see [9, Proposition 9.7]. The good-column hypothesis is used only to obtain the first normal-form board; subsequent iterations require only normal form.

Together, these results put a good-column Sudoku solution into a normal form, identify its exceptional q -adic rows, and allow those rows to be rescaled into a new Sudoku solution in normal form. Iterating this procedure reveals the successive values encoded by the decoration.

We next decorate each Sudoku line by a finite directed graph.

Let V be a finite set, and let $R \subseteq V^2$ be the edge set of a directed graph on V ; loops are allowed. For $\mu \in \mathcal{S}_q[N]$, let α_μ be its canonical affine function and let $\Gamma_\mu = \alpha_\mu^{-1}(0)$ be its canonical bad coset. If $\Gamma_\mu = \emptyset$, put $\text{Ref}(\mu) := \{\emptyset\}$. Otherwise set $L_\mu := q/2^{o_\mu}$, the step of the bad coset, and write $\Gamma_\mu = r + L_\mu\mathbb{Z}$. Then put

$$\text{Ref}(\mu) := \{(r + \ell L_\mu + qL_\mu\mathbb{Z}) \cap [N]: 0 \leq \ell < q\}.$$

This family is independent of the chosen representative r , since replacing r by $r + kL_\mu$ merely permutes the indices ℓ modulo q .

For a pair $(\mu, \chi) \in (C_q^\circ \times V)^{[N]}$, we call μ the *structural word* and regard χ as its decoration. Define $\mathcal{D}_R \subseteq (C_q^\circ \times V)^{[N]}$ by declaring $(\mu, \chi) \in \mathcal{D}_R$ when $\mu \in \mathcal{S}_q[N]$ and there are $(u, v) \in R$ and $\Delta \in \text{Ref}(\mu)$ such that the following conditions hold.

- (i) $\chi(n) = u$ for every $n \in [N] \setminus \Gamma_\mu$;
- (ii) $\chi(n) = v$ for every $n \in ([N] \cap \Gamma_\mu) \setminus \Delta$;
- (iii) no condition is imposed on $\chi|_\Delta$.

If $\Gamma_\mu = \emptyset$, this says that χ is the constant word u , with u the tail of an edge of R .

Lemma 3.1. *From finite lists encoding V and $R \subseteq V^2$, one can compute the complete list of elements of $\mathcal{D}_R$.*

Proof. We enumerate the fixed finite set $\mathcal{S}_q[N]$ and compute the canonical affine function, order, and bad coset of each word. First observe the following residue fact. If $q^3 \nmid t$ and $t' \equiv t \pmod{q^3}$, write $t = q^r u$ with $r \in \{0, 1, 2\}$ and $q \nmid u$. Then $t' = q^r(u + q^{3-r}z)$ for some $z \in \mathbb{Z}$; the factor in parentheses is again not divisible by q and is congruent to u modulo q . Hence $f_q(t') = f_q(t)$, so $f_q(t)$ is determined by $t \pmod{q^3}$ whenever $q^3 \nmid t$.

The case $a = b = 0$ gives the finitely many constant words obtained by reducing the odd factor c modulo q . Otherwise divide a common power of q out of a and b , and reduce c modulo q . We may therefore assume that a, b are not both divisible by q .

If $q \mid a$, then $q \nmid b$, so no $an + b$ is divisible by q , and the word is determined by $(a, b, c) \pmod{q}$. Suppose now that $q \nmid a$, and put $o = \nu_2(a) < s_0$. There is at most one $n \in [N]$ for which $q^3 \mid an + b$: if $n \neq n'$ were two such indices, then $2^{3s_0-o} \mid n - n'$, whereas $2^{3s_0-o} > q^2 > |n - n'|$. At every other index, the residue identity above shows that the digit is determined by $(a, b) \pmod{q^3}$.

For each pair $(\bar{a}, \bar{b}) \in C_{q^3}^2$ with $q \nmid \bar{a}$ and each odd $\bar{c} \in C_q$, test whether there is an index $n_0 \in [N]$ for which $q^3 \mid an_0 + b$. If no such index exists, the residue triple determines the word. If such an index exists, it is unique; retain an integral lift with $an_0 + b = q^3 z_0$. For each $\delta \in C_q^\circ$, replace b by $b + q^3(\tilde{\delta} - z_0)$, where $\tilde{\delta} \in \{1, \dots, q-1\}$ is the standard integer representative of δ . At n_0 this gives $f_q(an_0 + b) = \delta$, while at every other index the value is unchanged by the same residue identity. After multiplication by the odd coefficient c , these choices enumerate every possible nonzero digit at n_0 . The case $an_0 + b = 0$ requires no separate branch, because the convention $f_q(0) = 1$ is already among these nonzero choices before multiplication by c .

This finite loop lists every word in $\mathcal{S}_q[N]$ and retains, for each listed word, an explicit integer triple (a, b, c) realizing it. Before applying the Greenfeld–Tao normalization, we modify this representative so that $an + b \neq 0$ for every $n \in [N]$. Choose r so large that $q^r > \max_{n \in [N]} |an + b|$. Replacing b by $b + q^r$ leaves the word unchanged. Indeed, if $t = an + b \neq 0$, then $\nu_q(t) < r$ and $f_q(t + q^r) = f_q(t)$, while $f_q(q^r) = 1 = f_q(0)$. The new affine expression is nonzero as an integer at every $n \in [N]$. We may therefore absorb the odd coefficient c into the two affine coefficients, using $cf_q(t) = f_q(ct)$, and then divide out their common power of q .

The normalization in [9, Lemma 8.1] now computes the canonical order, canonical affine function, and its zero coset without requiring a separate case for an exact integer zero. These three objects depend only on the word, not on the retained realizing triple; see [9, Proposition 8.3]. In particular, the canonical bad coset is exactly the zero coset of the canonical affine function. The finite definitions of $\text{Ref}(\mu)$ and $\mathcal{D}_R$ then give an explicit computation of $\mathcal{D}_R$ from R . $\square$

The extraction of the word μ from a Sudoku line and the three regions of $[N]$ on which χ equals u , equals v , or is unrestricted are shown schematically in Figure 3.

A *line-decorated Sudoku pair* for R is a pair (Ξ, χ) , where $\Xi: \mathbb{B} \rightarrow C_q^\circ$ is a Sudoku solution and $\chi: \mathbb{B} \rightarrow V$, such that, for every $(i, j) \in \mathbb{Z}^2$, the map $[N] \rightarrow C_q^\circ \times V$ given by $n \mapsto (\Xi(n, jn + i), \chi(n, jn + i))$ belongs to $\mathcal{D}_R$. When R is fixed, we simply say that the pair is line-decorated. Such a pair is called *initially good* if Ξ also has good columns. The distinction matters: good columns are used only to reach the first normal form, and no good-column hypothesis is used thereafter.

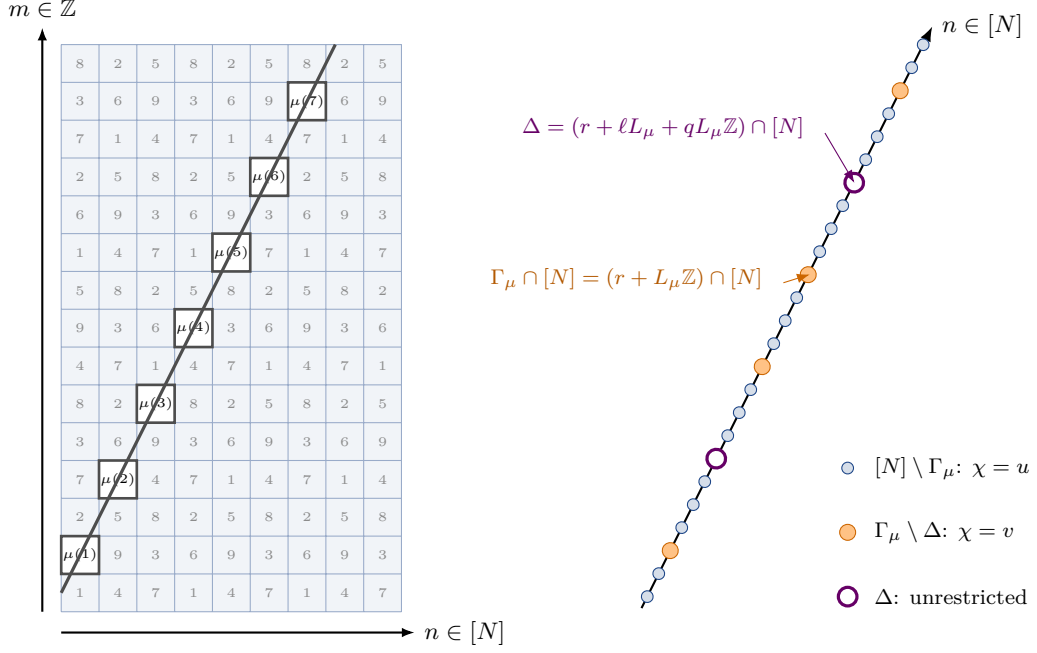


Figure 3: A Sudoku line and its decorated trace. The left panel selects the cells on $m = 2n + 1$; the right panel marks $[N] \setminus \Gamma_\mu$, $\Gamma_\mu \setminus \Delta$, and Δ , on which χ equals u , equals v , or is unrestricted, respectively. These marked sets are arithmetic progressions. The figure is schematic; the construction uses $q = 1024$.

To synchronize two decorated boards, put $U := [N] \times (\mathbb{Z} \setminus q\mathbb{Z})$ and call it the *normal layer* of a board. Let $R_1, R_2 \subseteq V^2$. A *bi-decorated solution* consists of two initially good boards $\Xi^1, \Xi^2: \mathbb{B} \rightarrow C_q^\circ$ and a common decoration $\Upsilon: \mathbb{B} \times \mathbb{B} \rightarrow V$ satisfying the following conditions.

- (i) For every $y \in \mathbb{B}$, the pair $(\Xi^1, \Upsilon(\cdot, y))$ is a line-decorated Sudoku pair for R_1 ;
- (ii) for every $x \in \mathbb{B}$, the pair $(\Xi^2, \Upsilon(x, \cdot))$ is a line-decorated Sudoku pair for R_2 .

Proposition 3.2. *A bi-decorated solution exists if and only if there is a map $T: \mathbb{N}^2 \rightarrow V$ such that, for every $(s, t) \in \mathbb{N}^2$, $(T(s, t), T(s + 1, t)) \in R_1$ and $(T(s, t), T(s, t + 1)) \in R_2$. Consequently, from a finite Wang instance one can compute the two finite relations $\mathcal{D}_{R_1}$ and $\mathcal{D}_{R_2}$, represented by their complete lists. A bi-decorated solution using these two relations and the fixed q -adic rules exists exactly when the Wang instance tiles $\mathbb{Z}^2$.*

Proof. We first establish the four one-board facts used in both directions. For these claims, fix an arbitrary directed relation $R \subseteq V^2$.

Claim 3.3. *If $\mathbf{p}: \mathbb{N} \rightarrow V$ satisfies $(\mathbf{p}(s), \mathbf{p}(s + 1)) \in R$ for every s , then an initially good line-decorated Sudoku pair exists.*

Proof. Let $\lambda(0) := 0$ and $\lambda(m) := \nu_q(m)$ for $m \neq 0$, and set $\Xi(n, m) := f_q(m)$ and $\chi(n, m) := \mathbf{p}(\lambda(m))$. The first map is the standard good-column solution of Greenfeld–Tao [9, Example 7.5]. On a line $m = jn + i$ with $(j, i) \neq (0, 0)$, let $s := \max\{r \in \mathbb{N}: q^r \mid j \text{ and } q^r \mid i\}$, which is finite because $(j, i) \neq (0, 0)$. Choose integers a, b with $(a, b) \notin q\mathbb{Z} \times q\mathbb{Z}$ such that $j = q^s a$ and $i = q^s b$. The definition $f_q(0) = 1$ and the identity $f_q(q^s t) = f_q(t)$ show that the structural word is the function $\mu(n) = f_q(an + b)$. Greenfeld–Tao [9, Lemma 8.1 and Proposition 8.3] show that its canonical bad coset is $\Gamma_\mu = \{n \in \mathbb{Z}: q \mid an + b\}$. Moreover, put $\Delta := \{n \in [N]: q^2 \mid an + b\}$. This set is empty when Γ_μ is empty and otherwise is one member of $\text{Ref}(\mu)$. To verify the latter assertion, suppose that $\Gamma_\mu \neq \emptyset$. Then $q \nmid a$. Put $o = \nu_2(a)$ and write $a = 2^o a_0$. Since the congruence $an + b \equiv 0 \pmod{q}$ has a solution, $2^o \mid b$; write $b = 2^o b_0$. Here a_0 is odd. The bad coset has step $L_\mu = q/2^o$, and $q^2 \mid an + b$ if and only if $a_0 n + b_0 \equiv 0 \pmod{qL_\mu}$. Since a_0 is odd, this is one residue class modulo qL_μ , hence one of the q refinements of Γ_μ . The decoration is $\mathbf{p}(s)$ outside the bad coset and $\mathbf{p}(s + 1)$ on the bad coset outside Δ . If $an + b = 0$ at some index, that index lies in Δ , where the decoration is unrestricted; thus the convention $\lambda(0) = 0$ causes no exceptional case.

Hence the edge $(\mathbf{p}(s), \mathbf{p}(s+1))$ witnesses membership in $\mathcal{D}_R$. The line $(j, i) = (0, 0)$ is constant and is witnessed by $(\mathbf{p}(0), \mathbf{p}(1))$. $\square$

The pair constructed in [Claim 3.3](#) has an additional useful feature: both fields are constant on every horizontal row. The structural entry removes the power of q dividing the row coordinate, whereas the decoration records that power as a position on the path $\mathbf{p}$; see [Figure 4](#).

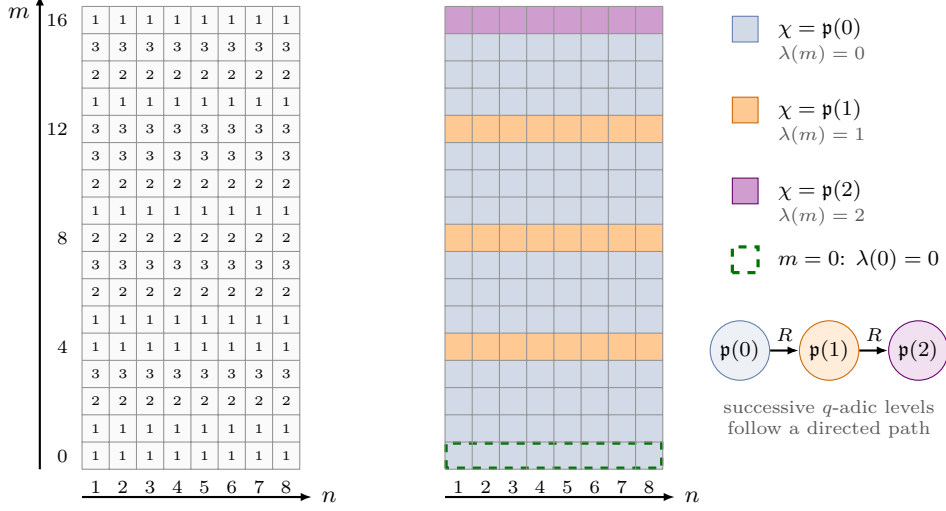


Figure 4: The standard decorated Sudoku pair from [Claim 3.3](#), shown for $q = 4$. Row m of the left board contains $f_4(m)$, while the corresponding row of the right board contains the path state $\mathbf{p}(\lambda(m))$; the arrows record consecutive path edges. Colors distinguish valuation levels, not necessarily distinct path vertices. The construction in the proof uses $q = 1024$.

Claim 3.4. *If (Ξ, χ) is line-decorated and Ξ is in normal form, there is one $T \in V$ such that $\chi(n, m) = T$ whenever $n \in [N]$ and $q \nmid m$.*

Proof. Fix a row index m with $q \nmid m$. On this row (3.2) gives the affine function

$$\alpha_m(n) := m + D \frac{q}{4} m(m - n) = m \left(1 + D \frac{q}{4} (m - n) \right) \pmod{q}.$$

Since $q/4$ is even, the factor in parentheses is odd for every $n \in \mathbb{Z}$. Multiplication by this factor does not change the 2-adic valuation of m , and therefore $\alpha_m(n) \neq 0$ in C_q for every $n \in \mathbb{Z}$.

Choose integer lifts of the two coefficients of α_m . The resulting integer affine form is never divisible by q , and hence its image under f_q agrees with α_m modulo q at every integer. Thus the row word has a representation whose bad coset is empty. By the uniqueness assertion of Greenfeld–Tao [9, Proposition 8.3], its canonical bad coset is empty. Consequently the decoration is constant on each such row.

We next identify the constants on rows having the same nonzero residue modulo q . Let $j \in q\mathbb{Z}$ and let i satisfy $q \nmid i$. Every point of the line $m = jn + i$ lies in the normal layer, and, because $j \equiv 0 \pmod{q}$, its structural word is $n \mapsto i + D \frac{q}{4} i(i - n) \pmod{q}$. Indeed, writing $j = q\ell$, we have

$$(jn + i)(jn + i - n) - i(i - n) = jn(jn + 2i - n).$$

The right-hand side is divisible by q , so the two terms remain congruent modulo q after multiplication by $q/4$. This affine function factors as $i(1 + D \frac{q}{4} (i - n))$ and is nonzero modulo q for every $n \in \mathbb{Z}$. Its canonical bad coset is therefore empty by the preceding representation argument, so the decoration is constant on this line. In particular, for $\rho \in \{1, \dots, q-1\}$ and $r \in \mathbb{Z}$, the slope- q line $m = qn + \rho + q(r-1)$ contains, at $n = 1$ and $n = 2$, points in the two rows $m = \rho + qr$ and $m = \rho + q(r+1)$. Hence the constants on those two rows agree. Iterating in r , all rows in a fixed nonzero residue class modulo q have one common decoration value.

It remains to compare the different nonzero residue classes. Let α_{diag} and Γ_{diag} denote, respectively, the canonical affine function and bad coset of the diagonal word $n \mapsto \Xi(n, n)$. By (3.2), $\Xi(n, n) = n \pmod{q}$ whenever $q \nmid n$. We claim that $\Gamma_{\text{diag}} = q\mathbb{Z}$.

First suppose that Γ_{diag} is a coset of $2\mathbb{Z}$. The step of α_{diag} is then an odd multiple of $q/2$; see [9, Lemma 8.1]. Hence α_{diag} is constant on each parity class. On the parity class complementary to Γ_{diag} we have $\Xi(n, n) = \alpha_{\text{diag}}(n)$. That class contains two elements of $[N]$ which are not divisible by q and are distinct modulo q —one may use 1, 3 in the odd case and 2, 4 in the even case, since the fixed q is at least 8. The preceding identity gives different values at these two points, a contradiction.

Thus Γ_{diag} is either empty or is a coset of $L\mathbb{Z}$ for some divisor L of q with $L \geq 4$. Among the q residue classes modulo q , the set $q\mathbb{Z}$ removes one even class, while Γ_{diag} , when nonempty, removes $q/L \leq q/4$ classes, all of the same parity. Since $q \geq 8$, there remain two residue classes of opposite parity outside $q\mathbb{Z} \cup \Gamma_{\text{diag}}$. Choose representatives $n_0, n_1 \in [N]$ of these classes. At both points, $\alpha_{\text{diag}}(n_i) = \Xi(n_i, n_i) = n_i \pmod{q}$. The difference $n_1 - n_0$ is odd and hence invertible modulo q , so these two equalities force $\alpha_{\text{diag}}(n) = n \pmod{q}$ for every $n \in \mathbb{Z}$. The canonical zero-set identity $\Gamma_{\text{diag}} = \alpha_{\text{diag}}^{-1}(0)$ in (i) above now gives $\Gamma_{\text{diag}} = q\mathbb{Z}$; in particular, the empty alternative is impossible.

Let $(u, v) \in R$ be an edge witnessing the decorated diagonal. Since the region outside the canonical bad coset is $[N] \setminus q\mathbb{Z}$, we have $\chi(n, n) = u$ whenever $q \nmid n$. This region meets every nonzero row residue modulo q . The row constants already shown to be constant within each such residue class must therefore all equal u . Taking $T = u$ proves the claim. $\square$

Claim 3.5. *If $(\mu, \chi) \in \mathcal{D}_R$ and B is an odd integer, then $(B\mu, \chi) \in \mathcal{D}_R$, with the same witnessing edge and refinement. Consequently, if (Ξ, χ) is line-decorated, then for arbitrary integers a, b, c with $a \neq 0$ and an odd integer B , define $\Xi'(n, m) := B\Xi(n, am + bn + c)$ and $\chi'(n, m) := \chi(n, am + bn + c)$. Then (Ξ', χ') is line-decorated. Here the coordinate map $(n, m) \mapsto (n, am + bn + c)$ maps $\mathbb{B} = [N] \times \mathbb{Z}$ into itself because it fixes the first coordinate.*

Proof. If $\mu(n) = c_0 f_q(an + b)$, then $B\mu(n) = (Bc_0) f_q(an + b)$ and Bc_0 is odd, including at an exact zero because the convention $f_q(0) = 1$ is multiplied by the same coefficient. Choose r so large that $q^r > \max_{n \in [N]} |an + b|$ and replace b by $b + q^r$. If $t = an + b \neq 0$, then $\nu_q(t) < r$ and $f_q(t + q^r) = f_q(t)$, while $f_q(q^r) = 1 = f_q(0)$. Thus this replacement removes every exact zero without changing either word. Absorb c_0 into the resulting affine expression for μ and Bc_0 into the one for $B\mu$. Since B is odd, the two normalized affine expressions differ by multiplication by the unit B , including after any common power of q is divided out. Greenfeld–Tao [9, Proposition 8.3] show that they are the canonical affine functions of μ and $B\mu$, respectively. Hence their orders, zero cosets, and families of q refinements agree, and the same decoration witness works. For the second assertion, a new line $m = jn + i$ maps to the old line $m = (aj + b)n + (ai + c)$ with the same n -parameter, so the first assertion applies line by line. $\square$

Claim 3.6. *Let (Ξ, χ) be a line-decorated pair with Ξ in normal form, and let T be its normal-layer value from Claim 3.4. Suppose that $A, C \in \mathbb{Z}$ and an odd integer B are such that the map Ξ^+ in (3.3) is again in normal form. Pull the decoration through the same coordinate map by setting $\chi^+(n, m) := \chi(n, q(m + An + C))$. If T^+ is the normal-layer value of χ^+ , then $(T, T^+) \in R$.*

Proof. By Claim 3.5, (Ξ^+, χ^+) remains line-decorated. On the old diagonal the bad-coset points in $[N]$ are $p_k := (qk, qk)$ for $1 \leq k \leq q$. The decorated-line witness is an edge $(u, v) \in R$. The points outside the bad coset lie in the normal layer, hence $u = T$. A refinement contains exactly one of the p_k , so $q - 1$ of them carry v .

Under this pullback, the preimage of p_k is $(qk, k - Aqk - C)$. Exactly one of these points lies on a new exceptional row, because its second coordinate is congruent to $k - C \pmod{q}$. Hence at least $q - 2 > 0$ indices avoid both the old refinement and the new exceptional row. For such an index, $v = \chi(p_k) = \chi^+(qk, k - Aqk - C) = T^+$. Thus the old witness edge is (T, T^+) . $\square$

These claims also show that an initially good line-decorated pair exists exactly when (V, R) has a right-infinite directed walk: Claim 3.3 gives one direction, while successive normalizations and Claims 3.4 to 3.6 give the other.

We now synchronize two copies of the one-board construction. Given T , use two standard boards and set $\Upsilon((n_1, m_1), (n_2, m_2)) := T(\lambda(m_1), \lambda(m_2))$. The proof of Claim 3.3 applies in each coordinate.

Conversely, normalize the two boards independently. Let Φ_0 and Ψ_0 be the affine coordinate maps used in these initial normalizations, and let $\Xi^{1,0}$ and $\Xi^{2,0}$ be the resulting Sudoku solutions in normal form. The accompanying odd scalar multipliers do not affect the decorations. At every subsequent

stage, choose a Tetris successor that is again in normal form. For $s, t \geq 0$, let Φ_s and Ψ_t denote the coordinate pullback maps from the stage- s and stage- t boards to the two original boards, respectively. Thus each board $\Xi^{1,s}$ or $\Xi^{2,t}$ is obtained from the corresponding original board by pullback along Φ_s or Ψ_t , followed by multiplication by an odd scalar. The successive Tetris coordinate maps have the form $\phi_s(n, m) := (n, q(m + A_s n + C_s))$ and $\psi_t(n, m) := (n, q(m + A'_t n + C'_t))$. We may take $\Phi_{s+1} := \Phi_s \circ \phi_s$ and $\Psi_{t+1} := \Psi_t \circ \psi_t$, while $\Xi^{1,s+1} := B_s \Xi^{1,s} \circ \phi_s$ and $\Xi^{2,t+1} := B'_t \Xi^{2,t} \circ \psi_t$ for odd integers B_s, B'_t .

Define $\Upsilon^{s,t}(x, y) := \Upsilon(\Phi_s(x), \Psi_t(y))$. Repeated application of [Claim 3.5](#) shows that, for every fixed y , the pair $(\Xi^{1,s}, \Upsilon^{s,t}(\cdot, y))$ is a line-decorated Sudoku pair for R_1 , and, for every fixed x , the pair $(\Xi^{2,t}, \Upsilon^{s,t}(x, \cdot))$ is a line-decorated Sudoku pair for R_2 . The odd scalars do not change any decoration witness, and the two coordinate-map families commute because they act on disjoint factors.

Applying [Claim 3.4](#) first in the first coordinate and then in the second shows that $\Upsilon^{s,t}$ is constant on $U \times U$; call its value $T(s, t)$. Fix $y \in U$. The decoration pulled through the first Tetris step is $x \mapsto \Upsilon^{s,t}(\phi_s(x), y) = \Upsilon^{s+1,t}(x, y)$. Hence [Claim 3.6](#) gives $(T(s, t), T(s+1, t)) \in R_1$. Similarly, fixing $x \in U$ and using $y \mapsto \Upsilon^{s,t}(x, \psi_t(y)) = \Upsilon^{s,t+1}(x, y)$ gives $(T(s, t), T(s, t+1)) \in R_2$. The two relations use the same value $T(s, t)$ because both are decoded from the same function $\Upsilon^{s,t}$ on $U \times U$.

[Figure 5](#) shows the Tetris pullback on two Sudoku windows. The filled points on the old board are precisely the displayed pairs that survive both exceptional sets. Repeating the same step independently in the two board coordinates produces the two-dimensional array in [Figure 6](#).

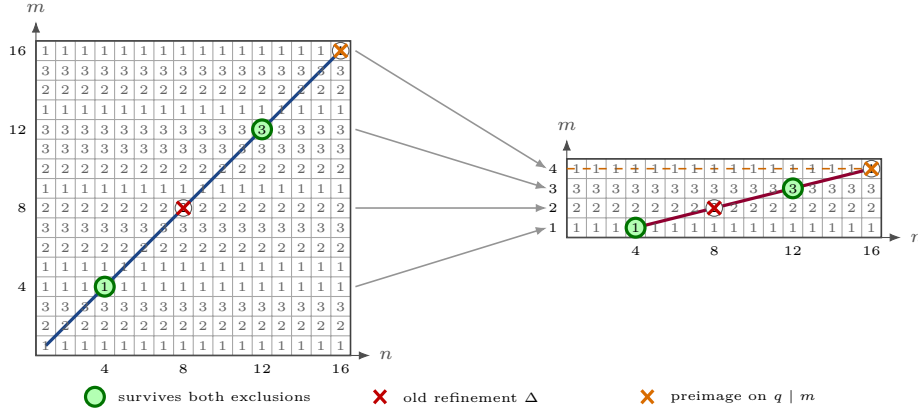


Figure 5: The Tetris pullback in the toy case $q = 4$, $A = C = 0$, and $B = 1$. Old row $4k$ feeds new row k , with corresponding marked points connected across the two boards. Green points avoid both excluded sets and yield $(T, T^+) \in R$; red points lie in the old refinement Δ , while orange points pull back to an exceptional row $q \mid m$. The proof uses $q = 1024$; nonzero A, C only shear and translate the preimage line.

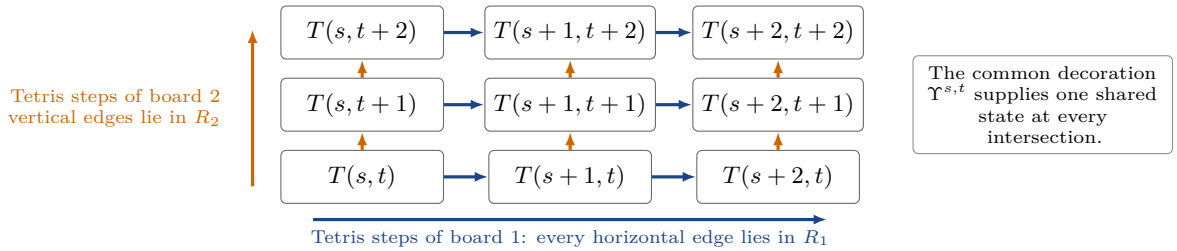


Figure 6: Independent iterations in the two board coordinates. Horizontal steps impose R_1 , vertical steps impose R_2 , and the common decoration supplies the shared state at each intersection. Only the coordinate pullbacks, not the relations, are asserted to commute.

We finally specialize the abstract relations to Wang tiles and verify that the required finite data are computable. Take the finite alphabet in the preceding construction to be the set $\mathcal{W}$ of distinct Wang tiles fixed in the introduction. Let $R_1, R_2 \subseteq \mathcal{W}^2$ express east–west and north–south color matching, respectively. A labeling $T: \mathbb{Z}^2 \rightarrow \mathcal{W}$ satisfying R_1 and R_2 is precisely a tiling by $\mathcal{W}$.

The two matching relations are visualized by the row and column paths in [Figure 1](#).

A legal tiling of $\mathbb{Z}^2$ restricts to $\mathbb{N}^2$. Conversely, suppose that $T: \mathbb{N}^2 \rightarrow \mathcal{W}$ satisfies the two matching relations. For each $k \geq 1$, translate T to a legal pattern on $[-k, k]^2$ by assigning $(x, y) \mapsto T(x + k, y + k)$. Extend this pattern by the fixed tile $T(0, 0)$ outside $[-k, k]^2$ to an element of $\mathcal{W}^{\mathbb{Z}^2}$. Since $\mathcal{W}$ is finite, the compact metrizable space $\mathcal{W}^{\mathbb{Z}^2}$ is sequentially compact, so there are indices $k_r \rightarrow \infty$ for which the corresponding extensions converge pointwise to some $\tilde{T} \in \mathcal{W}^{\mathbb{Z}^2}$. Every fixed adjacent pair of sites lies in $[-k_r, k_r]^2$ for all sufficiently large r . The two matching relations therefore pass to the limit, and $\tilde{T}$ is a legal Wang tiling of $\mathbb{Z}^2$. The relations R_1, R_2 are computable from $\mathcal{W}$, and Lemma 3.1 computes the corresponding finite decorated-line relations. These finite operations give the total computable map asserted in the proposition. $\square$

4 A fixed-degree finite-field state compiler

The input alphabet in Proposition 3.2 may be arbitrarily large. The purpose of this section is to show that an arbitrary pointwise relation of fixed arity can nevertheless be represented using finite abelian groups of uniformly bounded generator rank. The field order may grow; its extension degree remains fixed. We first construct the required fields effectively, then derive a field-rack normal form, and finally use one permutation row to exclude one prescribed state.

For a finitely generated abelian group L , let $\text{gen}(L)$ denote the least cardinality of a finite generating set, often called the *generator rank* when L is finite.

The following lemma gives the required finite-field construction.

Lemma 4.1. *Fix $m \geq 1$. There is an algorithm that, given a prime p , computes a presentation $\mathbb{K} = \mathbb{F}_p^m = \mathbb{F}_p[X]/(f)$, where f is monic irreducible of degree m , together with complete addition and multiplication tables and the polynomial basis $\eta_1 = 1, \eta_2 = \bar{X}, \dots, \eta_m = \bar{X}^{m-1}$. As an additive group, $\mathbb{K}^+ \cong C_p^m$ with this basis.*

Proof. Enumerate the monic polynomials f of degree m over $\mathbb{F}_p$ and test each one by the standard finite-field criterion $X^{p^m} \equiv X \pmod{f}$ and $\gcd(f, X^{p^{m/r}} - X) = 1$ for every prime r dividing m . For $m = 1$ the list of prime divisors $r \mid m$ is empty. All powers modulo f are computed by repeated squaring. These finite tests hold exactly when f is irreducible. A monic irreducible polynomial of every positive degree over a finite field exists, so the search terminates. Reduction modulo the first successful polynomial gives all field operations and the additive coordinates determined by this basis. $\square$

We next construct the *field rack*, an exact-cover system whose free parameter is an arbitrary field-valued state. Fix a prime p and an integer $m \geq 1$, and put $\mathbb{K} := \mathbb{F}_p^m$. Let U be a copy of the additive group $\mathbb{K}^+$. For the basis $\eta_1, \dots, \eta_m$, write $u_j \in U$ for the corresponding additive generator, and let $T := C_p t$ be a second auxiliary factor. For $r_1, \dots, r_m \in \mathbb{F}_p$, identify $r_1 u_1 + \dots + r_m u_m \in U$ with the field element $r_1 \eta_1 + \dots + r_m \eta_m$.

Let G be any finitely generated abelian group. Let $P, S: G \oplus U \oplus T \rightarrow \mathbb{K}$ be functions to be determined, and define $\Phi: G \oplus U \oplus T \rightarrow \mathbb{K}^2$ by $\Phi(x) := (P(x), S(x))$.

We first require that changing only the U -coordinate does not change S , and that changing only the T -coordinate does not change P . Since the u_j generate U and t generates T , it suffices to require, for every $x \in G \oplus U \oplus T$, that

$$S(x + u_j) = S(x) \quad \text{for every } j \in [m], \quad \text{and} \quad P(x + t) = P(x). \quad (4.1)$$

Thus $S(g + u + \tau t)$ is independent of u , whereas $P(g + u + \tau t)$ is independent of τ . Each of these equalities is enforced by a cylinder row from Lemma 2.2. The first family uses the second-coordinate quotient $\mathbb{K}^2 \rightarrow \mathbb{K}$, and the second uses the first-coordinate quotient.

For every $j \in [m]$, define the subgroup $L_j := \{(\eta_j a, -a) : a \in \mathbb{K}\} \leq \mathbb{K}^2$ and impose the fiberwise partition equation

$$(\Phi(x + u_j) + L_j) \uplus (\Phi(x + t) + (\mathbb{K}^2 \setminus L_j)) = \mathbb{K}^2. \quad (4.2)$$

Lemma 4.2. *The solutions of (4.1)–(4.2) are exactly the functions of the form $S(g + u + \tau t) = S(g) + \tau \zeta(g)$ and $P(g + u + \tau t) = P(g) + u \zeta(g)$, where $g \in G$, $u \in \mathbb{K}^+$, $\tau \in \mathbb{F}_p$, and $\zeta: G \rightarrow \mathbb{K}$ is arbitrary; the restrictions of P and S to G are also arbitrary. Here the product $u \zeta(g)$ is field multiplication. In particular, $\zeta(g) = \partial_t S(g)$ is an arbitrary $\mathbb{K}$ -valued field on G .*

Proof. The second set in (4.2) is the complement of one L_j -coset. Hence exact coverage is equivalent to $\Phi(x + u_j) + L_j = \Phi(x + t) + L_j$. Using (4.1), this is equivalent to $(\partial_{u_j} P(x), -\partial_t S(x)) \in L_j$, and therefore to $\partial_{u_j} P(x) = \eta_j \partial_t S(x)$. Put $\zeta(x) := \partial_t S(x)$. Since S is invariant under each u_j , commutation of differences gives $\zeta(x + u_j) = \zeta(x)$. Since P is t -periodic, the left side of the preceding identity is t -periodic. Multiplication by $\eta_j \neq 0$ is injective in the field, so $\zeta(x + t) = \zeta(x)$. Thus ζ descends to a function on G .

Integrating $\partial_t S = \zeta$ around the cyclic T -direction gives the stated formula for S ; it is consistent because $p\zeta(g) = 0$ in the additive group of $\mathbb{K}$. Integrating $\partial_{u_j} P = \eta_j \zeta$ in the basis directions gives the stated formula for P ; it is consistent because the increments commute and $p\eta_j \zeta(g) = 0$. The values $S(g)$ and $P(g)$ are unrestricted.

Conversely, functions of this form satisfy the invariance conditions (4.1) and the coupling identity above, hence every coupling row. No condition is imposed on ζ or on the restrictions of P and S to G beyond those stated. $\square$

We now implement a permutation test that excludes one field state. For $c \in \mathbb{K}$, impose

$$\bigsqcup_{r \in \mathbb{K}} (\Phi(x + r) + \{-rc\} \times \mathbb{K}) = \mathbb{K}^2, \quad (4.3)$$

where, in $\Phi(x + r)$, the symbol r denotes the element of U identified above with the field element $r \in \mathbb{K}$.

Lemma 4.3. *Assume the field-rack equations. The row (4.3) holds for every x if and only if $\zeta(g) \neq c$ for every $g \in G$, where $\zeta(g) = \partial_t S(g)$ is the state field from Lemma 4.2.*

Proof. Take $x = g + u + \tau t$. By the formula for P in Lemma 4.2, the summand of (4.3) indexed by $r \in \mathbb{K}$ is the vertical block $\{P(x) + r(\zeta(g) - c)\} \times \mathbb{K}$. These blocks partition $\mathbb{K}^2$ exactly when the values $P(x) + r(\zeta(g) - c)$, for $r \in \mathbb{K}$, run through $\mathbb{K}$ exactly once. This occurs precisely when multiplication by $\zeta(g) - c$ is a permutation of the field, or equivalently when $\zeta(g) - c \neq 0$. The criterion is independent of the unrestricted value $P(x)$ and of the auxiliary coordinates u and τ . $\square$

The field rack and the forbidden-state test are illustrated in Figure 7.

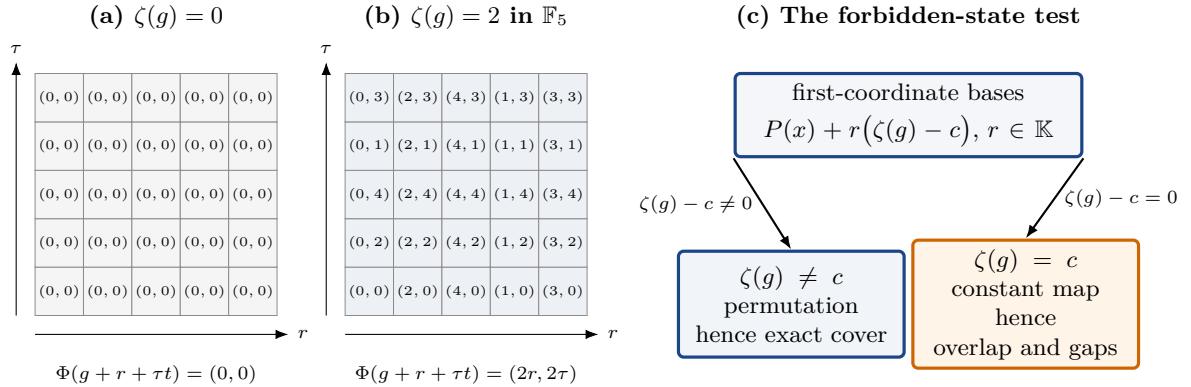


Figure 7: The field rack over $\mathbb{F}_5$ with the restrictions of P and S to G set to zero. In panels (a) and (b), each cell indexed by $(r, \tau) \in \mathbb{F}_5^2$ contains $\Phi(g + r + \tau t) = (r\zeta(g), \tau\zeta(g))$. When $\zeta(g) = 0$, all entries coincide; when $\zeta(g) = 2$, the entries run through $\mathbb{F}_5^2$. Panel (c) shows that the forbidden-state row partitions the first-coordinate fibers exactly when $r \mapsto r(\zeta(g) - c)$ is a permutation, equivalently when $\zeta(g) \neq c$.

The forbidden-state row gives the required compiler for every fixed arity.

Theorem 4.4. *Fix $m \geq 1$. Given a prime p , a finite relation $\Theta \subseteq \mathbb{F}_p^m$ specified by its complete list, and an encoding $G = \mathbb{Z}^r / L$ of a finitely generated abelian group in the format of Definition 2.1, one can effectively construct a finite fiberwise partition system whose only unknown is a function $\Phi: G \oplus \mathbb{F}_p^+ \oplus C_p t \rightarrow \mathbb{F}_p^2$. Write $\Phi = (P, S)$. After applying the coordinate isomorphism determined by the basis in Lemma 4.1 to $\partial_t S$, the resulting functions are exactly the maps $\vartheta: G \rightarrow \mathbb{F}_p^m$ satisfying $\vartheta(g) \in \Theta$ for every $g \in G$. The auxiliary domain factor $\mathbb{F}_p^+ \oplus C_p t$ has generator rank $m + 1$, and the value group $\mathbb{F}_p^2$ has generator rank $2m$.*

Proof. Construct $\mathbb{K} = \mathbb{F}_{p^m}$ as in Lemma 4.1, and let $\text{coord}: \mathbb{K} \rightarrow \mathbb{F}_p^m$ be the coordinate isomorphism determined by the chosen basis. Install the field rack and append the row (4.3) for every $c \in \mathbb{K} \setminus \text{coord}^{-1}(\Theta)$. By Lemma 4.3, the resulting system is equivalent to $\text{coord}(\zeta(g)) \in \Theta$ for every $g \in G$. Taking $\vartheta := \text{coord} \circ \zeta$ gives $\vartheta(g) \in \Theta$ for every $g \in G$. Conversely, given any such ϑ , set $\zeta := \text{coord}^{-1} \circ \vartheta$, choose arbitrary values of $P|_G$ and $S|_G$, and define P and S by the formulas in Lemma 4.2. If $\Theta = \emptyset$, the rows exclude every element of $\mathbb{K}$, so the system has no solution.

The construction is finite. The algorithm obtains the subgroups L_j , their complements, and every block in (4.3) by finite-field arithmetic and outputs them as complete element lists. Finally, $\mathbb{K}^+ \cong C_p^m$ and $(\mathbb{K}^2, +) \cong C_p^{2m}$, giving the stated generator bounds. $\square$

5 A bounded-generator fiberwise partition construction

We now combine the Sudoku encoding from Section 3 with the finite-field compiler from Section 4. We first fix and phase-pin the structural fiberwise partition systems, then encode the complete finite local state and identify its structural coordinates with the Sudoku fields. The field orders and the number of equations depend on the Wang instance, but the domain and value groups require uniformly bounded numbers of generators. The construction culminates in Theorem 5.5.

5.1 The phase-pinned structural equations

Put $\mathcal{A}_0 := \{0, 1\} \times \{0, \dots, s_0 - 1\} \times [N]$ and $R_0 := 1 + |\mathcal{A}_0| = 1 + 2s_0N$. Take $M_{\text{str}} := 26$, so that $2^{M_{\text{str}}} = 2^{26} > 2R_0 + 4$. Put $K_{\text{GT}} := (C_2)^3$ and $G_{\text{GT}} := \mathbb{Z}^2 \times K_{\text{GT}}$. Greenfeld–Tao define on G_{GT} a property S of the R_0 primary functions $\alpha_*: G_{\text{GT}} \rightarrow C_{2^{M_{\text{str}}}}$ and $\alpha_{a,b,n}: G_{\text{GT}} \rightarrow C_{2^{M_{\text{str}}}}$. Here α_* is the reference primary function, and the second family is indexed by $(a, b, n) \in \mathcal{A}_0$. Let e, e', e'' denote the three standard generators of the $(C_2)^3$ factor, with e the common Boolean flip. Write $\text{bin}_q(\xi_0, \dots, \xi_{s_0-1}) := \sum_{r=0}^{s_0-1} 2^r \xi_r \in C_q$ for binary decoding. We record the following two fixed finite relations so that the structural part of the local state relation used below can be checked directly. Let $\Omega \subseteq \{0, 1\}^{\mathcal{A}_0}$ consist of the tuples $\omega = (\omega_{a,b,n})$ such that

- (i) the word $n \mapsto \text{bin}_q(\omega_{1,0,n}, \dots, \omega_{1,s_0-1,n})$ lies in $\mathcal{S}_q[N]$;
- (ii) whenever $\text{bin}_q(\omega_{0,0,n}, \dots, \omega_{0,s_0-1,n}) \neq 0$, one has $\omega_{1,b,n} = \omega_{0,b,n}$ for $0 \leq b < s_0$.

Define its reflection-symmetric extension by

$$\tilde{\Omega} := \left\{ (\omega_*, \omega) \in \{0, 1\}^{1+|\mathcal{A}_0|} : (\omega_{a,b,n} \oplus \omega_*)_{(a,b,n) \in \mathcal{A}_0} \in \Omega \right\}.$$

These are the finite relations used in [9, equations (7.7)–(7.8)]; the two clauses defining Ω are properties I and II there. In particular, $\tilde{\Omega}$ is invariant under simultaneous complementation of all $1 + |\mathcal{A}_0|$ bits.

Following Greenfeld–Tao [9, Definitions 6.4 and 6.6], say that a primary tuple $\alpha = (\alpha_*, (\alpha_{a,b,n})_{(a,b,n) \in \mathcal{A}_0})$ obeys $P_{\tilde{\Omega}}$ if there are additive constants $a_j \in C_{2^{M_{\text{str}}}}$, a common odd unit $v \in C_{2^{M_{\text{str}}}}$, and Boolean functions $\kappa_j: G_{\text{GT}} \rightarrow \{0, 1\}$, indexed by $j \in \{*\} \sqcup \mathcal{A}_0$, such that the following conditions hold:

- (i) $\alpha_j(y) = a_j + v\kappa_j(y)$ for every $j \in \{*\} \sqcup \mathcal{A}_0$ and $y \in G_{\text{GT}}$;
- (ii) $\kappa_j(y + e) = 1 - \kappa_j(y)$ for every $j \in \{*\} \sqcup \mathcal{A}_0$ and $y \in G_{\text{GT}}$;
- (iii) $\kappa_j(y + e') = \kappa_j(y + e'') = \kappa_j(y)$ for every $j \in \{*\} \sqcup \mathcal{A}_0$ and $y \in G_{\text{GT}}$;
- (iv) $(\kappa_*(y), (\kappa_{a,b,n}(y))_{(a,b,n) \in \mathcal{A}_0}) \in \tilde{\Omega}$ for every $y \in G_{\text{GT}}$.

Here α_j denotes α_* for $j = *$ and $\alpha_{a,b,n}$ for $j = (a, b, n)$, and $0, 1 \in \{0, 1\}$ are identified with their natural images in $C_{2^{M_{\text{str}}}}$. Conditions (i)–(iii) give a common compatible Boolean normalization. The two possible common normalizations differ by simultaneous reflection, so the reflection symmetry of $\tilde{\Omega}$ makes $P_{\tilde{\Omega}}$ well defined.

We refer to the primary functions α_j before Boolean normalization as *raw tracks* when we need to distinguish them from the normalized Boolean functions κ_j .

The property S is defined by the following three axioms.

- (S1) The primary tuple satisfies $P_{\tilde{\Omega}}$.
- (S2) Each raw track $\alpha_{a,b,n}$ has period $(-n, 1)$.
- (S3) For every $n \in [N]$, the raw tuple $(\alpha_{0,b,n})_{0 \leq b < s_0}$ has the Boolean periodized-permutation property in direction $(1, 0)$ in the sense of Greenfeld–Tao [9, Definition 6.8].

The axiom (S3) includes compatible Booleanity and imposes the permutation condition after a common Boolean normalization. This direction has infinite order, so the shifts $j(1, 0)$ for $0 \leq j < 2^{s_0} = q$ are distinct, as required in the proof of [9, Proposition 6.9]. Greenfeld–Tao [9, Propositions 6.5, 6.7, and 6.9, Corollary 5.4, and Lemma 4.22] show that S is weakly expressible.

Here and below, the term *weakly expressible* is used in the exact existential-projection sense of Greenfeld–Tao; see [9, Section 4]. For every primary tuple α , the property $S(\alpha)$ holds precisely when a suitable finite-valued witness tuple makes an expressible extension hold. Thus the cited weak-expressibility results provide a finite witness index set $\mathcal{J}_{\text{wit}}$, finite abelian groups $(H_j)_{j \in \mathcal{J}_{\text{wit}}}$, and an expressible property S^* of the primary and witness functions such that, for every primary tuple α , one has $S(\alpha)$ if and only if $S^*(\alpha, \beta)$ holds for some witness tuple β . For definiteness, take S^* to be the witness-unrolled extension obtained by the following choices.

- (i) For each compatible Boolean constraint on L raw tracks, compare the last $L - 1$ tracks with the first by the two-track construction in [9, proof of Proposition 6.5].
- (ii) For every orbit under simultaneous complementation that is disjoint from $\tilde{\Omega}$, use the membership construction in [9, proof of Proposition 6.7].
- (iii) For each of the N Boolean periodized-permutation constraints, use the construction in [9, proof of Proposition 6.9].
- (iv) Retain the period constraints and, later, the reference pins as direct rows without introducing further value coordinates.

These choices fix all witness tracks and equations used below. Greenfeld–Tao [9, Equation (4.6)] then show that, after identifying each primary-and-witness tuple with one product-valued function, S^* is a finite system of fiberwise partition equations with value group

$$H_{\text{GT}} := C_{2^{M_{\text{str}}}}^{R_0} \times \prod_{j \in \mathcal{J}_{\text{wit}}} H_j.$$

Its distinguished primary quotient maps are the coordinate projections onto the R_0 displayed cyclic factors; taken jointly, these projections are onto $C_{2^{M_{\text{str}}}}^{R_0}$. Let δ_1, δ_2 denote the two free generators of G_{GT} , and define $S_{\text{ref}}(\alpha)$ to be the conjunction of $S(\alpha)$ with the requirement that, for every $x \in G_{\text{GT}}$,

$$\alpha_*(x + \delta_1) = \alpha_*(x) \quad \text{and} \quad \alpha_*(x + \delta_2) = \alpha_*(x). \quad (5.1)$$

To justify these phase-pinned equations, we prove the decoding implication used below rather than importing the unpinned converse of the published result. We first recall the corresponding condition for already normalized Boolean fields.

Let X be an abelian group and let $h \in X$. For $0 \leq b < s_0$, let $\xi_b: X \rightarrow \{0, 1\}$, and put $P(x) := \text{bin}_q(\xi_0(x), \dots, \xi_{s_0-1}(x)) \in C_q$. We call the tuple a *normalized Boolean periodized permutation* in direction h if $P(x + qh) = P(x)$ for every x and, for every x , the values $P(x + rh)$, $0 \leq r < q$, run through C_q exactly once. Simultaneously complementing all ξ_b replaces P by $q - 1 - P$, and therefore preserves this normalized property.

Proposition 5.1. *For the fixed q, N, M_{str} , there is a fixed finite abelian group H_{GT} with distinguished primary quotient maps to $C_{2^{M_{\text{str}}}}^{R_0}$, and there is a fixed finite fiberwise partition system $\mathcal{E}_{\text{GT}}$ on G_{GT} with the following properties.*

- (i) *The joint primary quotient map to $C_{2^{M_{\text{str}}}}^{R_0}$ is surjective.*
- (ii) *The projection of the solution set to the primary coordinates is exactly the set of tuples α satisfying $S_{\text{ref}}(\alpha)$.*

- (iii) Every solution decodes a uniquely determined good-column Sudoku board, and every such board, together with chosen witnessing column permutations, extends to a primary-and-witness solution of $\mathcal{E}_{\text{GT}}$.
- (iv) For each solution, XOR comparison with the normalized Boolean reference track defines semantic fields $\xi_{a,b,n}$ that are independent of the normalization. These fields are invariant under K_{GT} , and their role-zero restrictions $\xi_{a,b,n}$ perform the Sudoku decoding.

Proof. We first prove the decoding implication in (iii).

Claim 5.2. Suppose that bit fields $\xi_{a,b,n}: \mathbb{Z}^2 \rightarrow \{0,1\}$, indexed by $(a,b,n) \in \mathcal{A}_0$, satisfy the following conditions.

- (i) $(\xi_{a,b,n}(x))_{(a,b,n) \in \mathcal{A}_0} \in \Omega$ for every $x \in \mathbb{Z}^2$;
- (ii) $\xi_{a,b,n}$ has period $h_n = (-n, 1)$ for every (a,b,n) ;
- (iii) for every $n \in [N]$, the tuple $(\xi_{0,b,n})_{0 \leq b < s_0}$ is a normalized Boolean periodized permutation in direction $(1, 0)$.

Then the identities $\text{bin}_q(\xi_{a,0,n}(i,j), \dots, \xi_{a,s_0-1,n}(i,j)) = \Xi_a(n, jn+i)$ for $a \in \{0,1\}$, $n \in [N]$, and $i, j \in \mathbb{Z}$ define unique maps $\Xi_0, \Xi_1: \mathbb{B} \rightarrow C_q$. The map Ξ_1 is a Sudoku solution, and it has good columns witnessed by the permutations determined by Ξ_0 .

Proof. If $jn+i = j'n+i'$, then $(i'-i, j'-j) = (j'-j)(-n, 1)$, so condition (ii) makes the defining identity independent of the chosen pair (i, j) . Thus Ξ_0 and Ξ_1 are well defined and unique.

For fixed (i, j) , the first condition above and clause (i) in the definition of Ω show that $n \mapsto \Xi_1(n, jn+i)$ lies in $S_q[N]$; hence Ξ_1 is a Sudoku solution. Clause (ii) gives $\Xi_0(n, m) \neq 0 \implies \Xi_1(n, m) = \Xi_0(n, m)$. For fixed n , condition (iii) says that the decoded function $m \mapsto \Xi_0(n, m)$ is q -periodic and assumes every value of C_q exactly once in a period. It therefore has the form $\Xi_0(n, m) = \sigma_n(m \bmod q)$ for a permutation $\sigma_n: C_q \rightarrow C_q$. Whenever the right-hand side is nonzero, the preceding implication gives $\Xi_1(n, m) = \sigma_n(m \bmod q)$, which is exactly the good-column condition. $\square$

We now establish (i), (ii), the lifting assertion in (iii), and (iv). Let Pin_{ref} denote the two period conditions in (5.1). Its expressibility follows directly from two applications of Lemma 2.2; see also [9, Corollary 5.4]. Lift it to the full primary-and-witness tuple and put $S_{\text{ref}}^*(\alpha, \beta) := S^*(\alpha, \beta) \wedge \text{Pin}_{\text{ref}}(\alpha)$. Greenfeld–Tao [9, Lemma 4.22(i),(iii)] show that this property is expressible, and

$$\exists \beta S_{\text{ref}}^*(\alpha, \beta) \iff \text{Pin}_{\text{ref}}(\alpha) \wedge \exists \beta S^*(\alpha, \beta) \iff S_{\text{ref}}(\alpha). \quad (5.2)$$

Use the witness-unrolled expressible extension fixed above, together with an invariant-factor presentation of its value group and explicit lists for all blocks and shifts. The product form of H_{GT} makes the joint primary quotient map surjective. These choices define the finite system $\mathcal{E}_{\text{GT}}$ in the format of Definition 2.1; they are independent of the Wang instance and form a fixed part of the construction.

To prove the extension assertion in part (iii), start with a good-column board and chosen column permutations. In the construction of [9, equations (7.5)–(7.8)], the reference track is $\alpha_*(x, (\varepsilon, \varepsilon', \varepsilon'')) = \varepsilon$. It is invariant in both free directions, so the complete witness tuple supplied by the weak-expressibility construction solves S_{ref}^* .

For the converse, axiom (S1), namely P_{Ω} , includes compatible Booleanity. Its defining normalization, equivalently Greenfeld–Tao [9, equation (6.7)], gives additive constants $a_j \in C_{2^{M_{\text{str}}}}$, one odd unit $v \in C_{2^{M_{\text{str}}}}$, and normalized Boolean tracks $\kappa_j: G_{\text{GT}} \rightarrow \{0,1\}$, indexed by $j \in \{*\} \sqcup \mathcal{A}_0$, such that

$$\alpha_j = a_j + v\kappa_j \quad \text{for every } j \in \{*\} \sqcup \mathcal{A}_0. \quad (5.3)$$

Here α_j denotes α_* for $j = *$ and $\alpha_{a,b,n}$ for $j = (a, b, n)$, and $0, 1 \in \{0,1\}$ are identified with their natural images in $C_{2^{M_{\text{str}}}}$. The only ambiguity in these normalization parameters is the simultaneous reflection obtained, for every $j \in \{*\} \sqcup \mathcal{A}_0$, from

$$v' = -v, \quad a'_j = a_j + v, \quad \kappa'_j = 1 - \kappa_j. \quad (5.4)$$

Indeed, $a'_j + v'\kappa'_j = (a_j + v) - v(1 - \kappa_j) = a_j + v\kappa_j$, so the simultaneous change of v , all a_j , and all κ_j leaves every raw primary track unchanged.

Because v is odd, it is a unit of $C_{2^{\mathcal{M}_{\text{str}}}}$. Compatible Booleanity gives, for every $j \in \{*\} \sqcup \mathcal{A}_0$ and every $y \in G_{\text{GT}}$, $\kappa_j(y + e) = 1 - \kappa_j(y)$ and $\kappa_j(y + e') = \kappa_j(y + e'') = \kappa_j(y)$. Applying (5.1) to (5.3) gives $\kappa_*(x + \delta_r, \eta) = \kappa_*(x, \eta)$ for $r = 1, 2$, $x \in \mathbb{Z}^2$, and $\eta \in K_{\text{GT}}$. Hence $c := \kappa_*(x, 0) \in \{0, 1\}$ is independent of $x \in \mathbb{Z}^2$. For every $(a, b, n) \in \mathcal{A}_0$, define the full-role semantic field $\tilde{\xi}_{a,b,n}: G_{\text{GT}} \rightarrow \{0, 1\}$ and its role-zero restriction $\xi_{a,b,n}: \mathbb{Z}^2 \rightarrow \{0, 1\}$ by

$$\tilde{\xi}_{a,b,n}(y) := \kappa_{a,b,n}(y) \oplus \kappa_*(y), \quad \xi_{a,b,n}(x) := \tilde{\xi}_{a,b,n}(x, 0) = \kappa_{a,b,n}(x, 0) \oplus c. \quad (5.5)$$

Both normalized tracks complement under e and both are invariant under e', e'' . Thus every $\tilde{\xi}_{a,b,n}$ is invariant under all of K_{GT} . The full reflection (5.4) also complements both arguments of the XOR, so it leaves both $\tilde{\xi}_{a,b,n}$ and $\xi_{a,b,n}$ unchanged.

The $\tilde{\Omega}$ part of $P_{\tilde{\Omega}}$ says precisely that $(\tilde{\xi}_{a,b,n}(y))_{(a,b,n) \in \mathcal{A}_0}$ lies in Ω for every $y \in G_{\text{GT}}$, and hence that $(\xi_{a,b,n}(x))_{(a,b,n) \in \mathcal{A}_0}$ lies in Ω for every $x \in \mathbb{Z}^2$. If $h_n = (-n, 1)$, axiom (S2) gives the period h_n to every raw track with index n . Since v is a unit, the same period holds for $\kappa_{a,b,n}$. The reference pins make κ_* invariant under both free directions and hence give it period h_n . Therefore both $\tilde{\xi}_{a,b,n}$ and its role-zero restriction $\xi_{a,b,n}$ have period h_n . For fixed n , uniqueness up to simultaneous reflection allows the common normalization in axiom (S3) to be taken as the restriction of (5.3). Hence $(\kappa_{0,b,n}(\cdot, 0))_{0 \leq b < s_0}$ is a normalized Boolean periodized permutation in direction $(1, 0)$ in the sense defined above. The role-zero relative tuple is either this tuple or its simultaneous complement. The final sentence of the terminology paragraph above therefore shows that it has the same property.

The fields $\xi_{a,b,n}$ now satisfy conditions (i)–(iii) of Claim 5.2. That claim defines a unique good-column Sudoku board from these fields. Since the fields in (5.5) are unchanged by the simultaneous reflection (5.4), this decoded board is uniquely determined by the primary tuple. $\square$

The argument following [9, equations (7.7)–(7.8)] forms relative Boolean fields by comparing every nonreference primary track with the reference track and then infers skew periods and the periodized-permutation property from axioms imposed on the raw nonreference primary tracks. Because the reference track also enters the relative fields, that inference requires control of its free-coordinate phase. The following diagnostic example isolates the precise failure.

There is a primary tuple satisfying (S1)–(S3) for the unpinned property S for which the relative bits defined by comparison with the reference track fail both a required skew period and the normalized Boolean periodized-permutation condition. To construct one, use the standard board $\Xi(n, m) = f_q(m)$ and the identity column permutations. Let $\xi_{a,b,n}^\circ: \mathbb{Z}^2 \rightarrow \{0, 1\}$ be the Boolean functions of Greenfeld–Tao [9, equations (7.5)–(7.6)]. The normalized primary tuple has

$$\kappa_*(x, (\varepsilon, \varepsilon', \varepsilon'')) = \varepsilon, \quad \kappa_{a,b,n}(x, (\varepsilon, \varepsilon', \varepsilon'')) = \xi_{a,b,n}^\circ(x) \oplus \varepsilon.$$

Choose additive constants a_* and $(a_{a,b,n})_{(a,b,n) \in \mathcal{A}_0}$ and a common odd unit v , and define the raw primary tuple α from the displayed normalized functions by (5.3). Put $x_0 := (1, 0)$ and replace the reference track at every point whose free-coordinate component is x_0 by

$$\kappa'_*(x, (\varepsilon, \varepsilon', \varepsilon'')) := \begin{cases} 1 - \varepsilon, & x = x_0, \\ \varepsilon, & x \neq x_0. \end{cases}$$

Define the corresponding modified raw primary tuple by $\alpha'_* := a_* + v\kappa'_*$ and $\alpha'_{a,b,n} := \alpha_{a,b,n}$ for every $(a, b, n) \in \mathcal{A}_0$. The new reference track remains Boolean in the reference flip and invariant under the other two role flips, with the same additive constant a_* and common odd unit v as before.

The preceding paragraph shows that the compatible Boolean part of (S1) remains valid. Away from x_0 , its normalized $\tilde{\Omega}$ constraint is unchanged. At x_0 , the two encoded words indexed by $a = 0, 1$ in the standard tuple are both the constant word with binary value 1. Complementing the reference complements every relative bit, so both words become the constant word with binary value $(q - 1) - 1 = q - 2$. This belongs to $S_q[N]$: in (3.1), take $a = 0$, $b = q - 2$, and $c = 1$. It is nonzero, and the two encoded words still agree. Thus the normalized pointwise constraint, and hence (S1), remains valid. Axioms (S2) and (S3) use only

the unchanged nonreference primary tracks $\alpha'_{a,b,n} = \alpha_{a,b,n}$, so they also remain valid. Thus the modified raw primary tuple still satisfies (S1)–(S3).

For any $n \in [N]$, the points x_0 and $x_0 + (-n, 1)$ represent the same Sudoku cell. Each of the two relative words at the first point now encodes $q - 2$, whereas its counterpart at the second point still encodes 1, so the required relative skew period fails. Moreover, in the length- q window beginning at x_0 in direction $(1, 0)$, the $a = 0$ relative word takes the value $q - 2$ both at x_0 and at the unchanged point $(q - 2, 0)$. Hence that relative word is not a periodized permutation. Thus the raw axioms alone do not justify the stated relative-field inference. The two pins in (5.1) remove this freedom without selecting an absolute Boolean phase.

This issue concerns only the relative-field inference from the unpinned raw axioms, not the main theorem of Greenfeld–Tao [9]. The forward construction from a good-column board has a reference track that is invariant under both free directions, and the present paper uses only the explicitly pinned property S_{ref} and the decoding established in Proposition 5.1.

More precisely, by the same projection argument as in (5.2), if $C(\alpha, \theta)$ is any later constraint that does not involve the structural witness tuple β , then

$$\exists \beta (S_{\text{ref}}^*(\alpha, \beta) \wedge C(\alpha, \theta)) \iff S_{\text{ref}}(\alpha) \wedge C(\alpha, \theta).$$

We use this exact projection identity for primary pins, mixed bridges, and the finite local state relation.

For $\ell = 1, 2$, let $v_{\ell,1}, v_{\ell,2}$ be the free generators of a copy $\mathbb{Z}_{(\ell)}^2$, and let $e_\ell, e'_\ell, e''_\ell$ be the three distinguished generators of $K_{\text{GT}}^{(\ell)}$; set $E_\ell := \{e_\ell, e'_\ell, e''_\ell\}$. Regard two disjoint copies of G_{GT} as direct factors of $G_{\text{str}} := \mathbb{Z}^4 \times K_{\text{GT}}^{(1)} \times K_{\text{GT}}^{(2)}$. Install the two copies of $\mathcal{E}_{\text{GT}}$ on G_{str} by embedding their shifts in the respective copies of G_{GT} , as in Greenfeld–Tao; see [9, Definition 4.20 and Lemma 4.22(ii)]. Rename their witness index sets disjointly and cylinder-lift both systems to $H_{\text{str}} := H_{\text{GT}}^{(1)} \times H_{\text{GT}}^{(2)}$. Thus the restriction to every coset of either embedded subgroup $G_{\text{GT}}^{(\ell)}$ satisfies the original system, and the components of both complete primary-and-witness tuples are coordinates of a single product-valued function. A *primary quotient* means one of the fixed maps $H_{\text{str}} \rightarrow C_{2^{M_{\text{str}}}}$ belonging to those primary coordinates. In the installed system, a raw primary track is the composition of one of these quotients with the product-valued function. Witness coordinates are never read by a later bridge or by the local state relation. In completeness constructions, the complete primary-and-witness tuple of copy ℓ is extended constantly over the other structural copy and, in the full assembly below, over the auxiliary finite-field domain factor. In soundness, these additional periods are imposed only on the raw primary tracks.

Thus $K_{\text{GT}}^{(1)} \times K_{\text{GT}}^{(2)}$ is the previously announced group $K_{\text{str}} = (C_2)^6$. The groups K_{str} and H_{str} , their presentations, their primary quotient maps, and the two installed fiberwise partition systems are now fixed independently of the Wang instance.

We call $\mathbb{Z}^4 \times \{0\} \leq G_{\text{str}}$ the *role-zero subgroup*. The auxiliary finite-field directions introduced below are separate from this terminology.

5.2 Assembly of the fiberwise partition system

We begin with an exact-cover identity that will identify a structural Boolean field with one coordinate of the finite-field state.

Let H be a finite abelian group, let $\varphi = (\rho_A, \rho_B): H \twoheadrightarrow A \times B$ be a jointly surjective quotient, and let $\alpha: G \rightarrow H$. For $C \subseteq A \times B$ and $y \in G$, write $C @ y := \alpha(y) + \varphi^{-1}(C)$. Put $A^* := A \setminus \{0\}$, $B^* := B \setminus \{0\}$, and define $\mathcal{B}_{10} := A^* \times \{0\}$, $\mathcal{B}_{01} := \{0\} \times B^*$, $\mathcal{B}_{11} := A^* \times B^*$, and $\mathcal{B}_{00} := \{(0, 0)\}$.

Lemma 5.3. *Suppose that $\rho_A \alpha(y + s) = \rho_A \alpha(y)$ and $\rho_B \alpha(y + r) = \rho_B \alpha(y)$ for every $y \in G$. Fix $c \in B$, and put $X := \partial_r(\rho_A \alpha)(x)$ and $Z := \partial_s(\rho_B \alpha)(x)$. Then the fiberwise partition equation*

$$(\mathcal{B}_{10} @ x) \uplus (\mathcal{B}_{11} @ (x + r)) \uplus ((\mathcal{B}_{00} - (0, c)) @ (x + s)) \uplus ((\mathcal{B}_{01} - (0, c)) @ (x + r + s)) = H$$

holds if and only if $X = 0$ or $Z = c$.

Proof. Using the notation $C @ y$ introduced above, translate the four sets by $-\alpha(x)$ and pass to the quotient $A \times B$. The two cross-period identities in the hypothesis give the four quotient blocks $\mathcal{B}_{10}$, $(X, 0) + \mathcal{B}_{11}$, $(0, Z - c) + \mathcal{B}_{00}$, and $(X, Z - c) + \mathcal{B}_{01}$. If $X = 0$, the two quotient blocks meeting each

vertical line partition that line; if $Z = c$, the analogous statement holds for each horizontal line. If $X \neq 0$ and $Z \neq c$, the quotient point $(X, 0)$ lies both in the translated $\mathcal{B}_{10}$ block and in the translated $\mathcal{B}_{01}$ block, because $c - Z \in B^*$. Since φ is surjective, this quotient overlap has a preimage in H , so the cover is not exact. Thus the asserted fiberwise partition equation is equivalent to the asserted disjunction. $\square$

We now define the finite local state that will be stored in one field element. Let a nonempty Wang instance $\mathcal{W}$ be given, and let $R_1, R_2 \subseteq \mathcal{W}^2$ be the east–west and north–south matching relations. Recall that $\mathcal{A}_0 = \{0, 1\} \times \{0, \dots, s_0 - 1\} \times [N]$. Define the fixed index sets

$$\mathcal{I}_{\text{sem}} := \{1, 2\} \times \mathcal{A}_0, \quad \mathcal{I}_{\text{dec}} := [N] \times [N], \quad \mathcal{I}_0 := \mathcal{I}_{\text{sem}} \sqcup \mathcal{I}_{\text{dec}}.$$

Set $m_0 := |\mathcal{I}_0| = 2|\mathcal{A}_0| + N^2 = 4s_0N + N^2$. For $j = (\ell, a, b, n) \in \mathcal{I}_{\text{sem}}$, write $\tilde{\xi}_j := \tilde{\xi}_{a,b,n}^\ell$ and $\xi_j := \xi_{a,b,n}^\ell$.

Let $p_{\mathcal{W}}$ be the least odd prime larger than $|\mathcal{W}|$ and choose, in the input order, a computable injection $\text{code}_{\mathcal{W}}: \mathcal{W} \hookrightarrow \mathbb{F}_{p_{\mathcal{W}}}$. Extend the inverse of $\text{code}_{\mathcal{W}}$ to a total map $\text{dec}_{\mathcal{W}}: \mathbb{F}_{p_{\mathcal{W}}} \rightarrow \mathcal{W}$ by sending values outside the code image to the first tile of $\mathcal{W}$.

For $\mathbf{b} = (b_j)_{j \in \mathcal{I}_{\text{sem}}}$ and $\omega = (\omega_{n_1, n_2})_{(n_1, n_2) \in \mathcal{I}_{\text{dec}}}$ in $\mathbb{F}_{p_{\mathcal{W}}}^{\mathcal{I}_0}$, whenever the structural coordinates are bits, define $\mu_{\mathbf{b}}^\ell(n) := \text{bin}_q(b_{\ell, 1, 0, n}, \dots, b_{\ell, 1, s_0 - 1, n}) \in C_q$ for $\ell = 1, 2$, and put $\chi_\omega(n_1, n_2) := \text{dec}_{\mathcal{W}}(\omega_{n_1, n_2})$. Here $0, 1 \in \mathbb{F}_{p_{\mathcal{W}}}$ are identified with the corresponding integers in the binary decoding map bin_q .

Using these two definitions, let $\Theta_{\mathcal{W}} \subseteq \mathbb{F}_{p_{\mathcal{W}}}^{\mathcal{I}_0}$ consist of the assignments $(\mathbf{b}, \omega)$ satisfying the following conditions.

- (a) for $\ell = 1, 2$, all coordinates $b_{\ell, a, r, n}$ lie in $\{0, 1\} \subseteq \mathbb{F}_{p_{\mathcal{W}}}$, and the resulting bit tuple $(b_{\ell, a, r, n})_{(a, r, n) \in \mathcal{A}_0}$ belongs to Ω ;
- (b) every ω_{n_1, n_2} belongs to $\text{im}(\text{code}_{\mathcal{W}})$;
- (c) for every $n_2 \in [N]$, the map $[N] \rightarrow C_q^\circ \times \mathcal{W}$ given by $n_1 \mapsto (\mu_{\mathbf{b}}^1(n_1), \chi_\omega(n_1, n_2))$ belongs to $\mathcal{D}_{R_1}$;
- (d) for every $n_1 \in [N]$, the map $[N] \rightarrow C_q^\circ \times \mathcal{W}$ given by $n_2 \mapsto (\mu_{\mathbf{b}}^2(n_2), \chi_\omega(n_1, n_2))$ belongs to $\mathcal{D}_{R_2}$.

The relation $\Theta_{\mathcal{W}}$ is explicitly computable. Indeed, its arity m_0 is fixed, all ambient sets are finite, and the decorated-line relation $\mathcal{D}_R$ is computable from its finite alphabet and edge relation by [Lemma 3.1](#).

For the fixed integer m_0 , construct $\mathbb{K}_{\mathcal{W}} := \mathbb{F}_{p_{\mathcal{W}}}^{m_0}$ using [Lemma 4.1](#). Index its chosen $\mathbb{F}_{p_{\mathcal{W}}}$ -basis by $\mathcal{I}_0$, write $\text{coord}_{\mathcal{W}}: \mathbb{K}_{\mathcal{W}} \rightarrow \mathbb{F}_{p_{\mathcal{W}}}^{\mathcal{I}_0}$ for the corresponding coordinate isomorphism, and let $\lambda_j: \mathbb{K}_{\mathcal{W}} \rightarrow \mathbb{F}_{p_{\mathcal{W}}}$, for $j \in \mathcal{I}_0$, be its coordinate maps.

We next assemble the domain and value groups and specify the finite pin list. Let $\mathcal{R}_{\mathcal{W}} := \mathbb{K}_{\mathcal{W}}^+ \oplus C_{p_{\mathcal{W}}} t$. For $j \in \mathcal{I}_0$, let u_j denote the additive basis generator corresponding to the basis element η_j of the field. Set

$$G_{\mathcal{W}} := G_{\text{str}} \times \mathcal{R}_{\mathcal{W}}, \quad H_{\mathcal{W}} := H_{\text{str}} \times \mathbb{K}_{\mathcal{W}}^2.$$

For these groups, write $\alpha: G_{\mathcal{W}} \rightarrow H_{\mathcal{W}}$ for the common product-valued map and write $P, S: G_{\mathcal{W}} \rightarrow \mathbb{K}_{\mathcal{W}}$ for the two coordinate functions of its field projection. Every field row below is cylinder-lifted through the quotient $H_{\mathcal{W}} \twoheadrightarrow \mathbb{K}_{\mathcal{W}}^2$. Write $\text{pr}_S: H_{\mathcal{W}} \rightarrow \mathbb{K}_{\mathcal{W}}$ for the second field-coordinate quotient. For $\ell = 1, 2$, let ρ_*^ℓ and $\rho_{a,b,n}^\ell$ denote the primary quotient maps $H_{\mathcal{W}} \twoheadrightarrow C_{2M_{\text{str}}}$ obtained from the reference track and the indexed primary tracks of structural copy ℓ , respectively. We call $\rho_*^\ell \alpha$ and $(\rho_{a,b,n}^\ell \alpha)_{(a,b,n) \in \mathcal{A}_0}$ the *primary data* of copy ℓ . Install the field-rack rows of [Lemma 4.2](#) and append the forbidden-state row for every $c \in \mathbb{K}_{\mathcal{W}} \setminus \text{coord}_{\mathcal{W}}^{-1}(\Theta_{\mathcal{W}})$. By [Lemma 4.2](#), $\partial_t S$ descends to the state field $\zeta: G_{\text{str}} \rightarrow \mathbb{K}_{\mathcal{W}}$.

Each structural fiberwise partition system is installed on $G_{\mathcal{W}}$ by embedding its shifts in the corresponding copy of G_{GT} and cylinder-lifting its blocks through $H_{\mathcal{W}} \twoheadrightarrow H_{\text{str}}$. Thus the restriction to every coset of the corresponding subgroup $G_{\text{GT}}^{(\ell)}$ satisfies the original fixed system. Relative to copy ℓ , we call $G_{\text{GT}}^{(3-\ell)} \times \mathcal{R}_{\mathcal{W}}$ its *transverse subgroup*.

If $\rho: H_{\mathcal{W}} \twoheadrightarrow L$ is a named quotient and $h \in G_{\mathcal{W}}$, write $\text{Per}(\rho, h)$ for the cylinder row equivalent to $\partial_h(\rho \alpha) = 0$. For $\ell = 1, 2$, put $\mathcal{T}_\ell := \{v_{3-\ell, 1}, v_{3-\ell, 2}\} \cup E_{3-\ell} \cup \{u_j: j \in \mathcal{I}_0\} \cup \{t\}$. With these transverse sets, append the following finite rows, each period row being of the form specified above.

- (P1) By (5.1), the reference track of copy ℓ already has periods $v_{\ell, 1}$ and $v_{\ell, 2}$. Add $\text{Per}(\rho_*^\ell, h)$ for every $h \in \mathcal{T}_\ell$.
- (P2) For each ℓ , every $(a, b, n) \in \mathcal{A}_0$, and every $h \in \mathcal{T}_\ell$, add $\text{Per}(\rho_{a,b,n}^\ell, h)$.

- (P3) For each decoration index $(n_1, n_2) \in [N]^2$, put $V_{n_1, n_2} := \lambda_{n_1, n_2} \circ \text{pr}_S: H_{\mathcal{W}} \rightarrow \mathbb{F}_{p_{\mathcal{W}}}$ and add both $\text{Per}(V_{n_1, n_2}, -n_1 v_{1,1} + v_{1,2})$ and $\text{Per}(V_{n_1, n_2}, -n_2 v_{2,1} + v_{2,2})$.
- (P4) For every $h \in E_1 \cup E_2$, add $\text{Per}(\text{pr}_S, h)$.

Witness coordinates of the fixed structural systems are not pinned.

For $j = (\ell, a, b, n) \in \mathcal{I}_{\text{sem}}$, define $U_j^- := \rho_{a,b,n}^\ell - \rho_*^\ell$ and $U_j^+ := \rho_{a,b,n}^\ell + \rho_*^\ell$.

Claim 5.4. *Every solution of P1–P4 has the following properties.*

- (i) For $\ell = 1, 2$, the primary data of structural copy ℓ are invariant under its transverse subgroup.
- (ii) For every structural index j belonging to copy ℓ , the semantic field $\tilde{\xi}_j$ is invariant under $G_{\text{GT}}^{(3-\ell)} \times K_{\text{GT}}^{(\ell)} \times \mathcal{R}_{\mathcal{W}}$.
- (iii) The state field $\zeta = \partial_t S$ is invariant under K_{str} and hence descends, together with its coordinates, to $\mathbb{Z}^4$.
- (iv) For every structural index j belonging to copy ℓ , the quotient $U_j^\pm$ has period t , while $\lambda_j \circ \text{pr}_S$ has period e_ℓ .
- (v) For every $(n_1, n_2) \in [N]^2$, writing a point of $\mathbb{Z}^4$ as $x = ((i_1, j_1), (i_2, j_2))$, the decoration coordinate $\lambda_{n_1, n_2}(\zeta(x))$ depends on x only through $(j_1 n_1 + i_1, j_2 n_2 + i_2)$.

Proof. P1–P2 give the stated periods for every raw primary track on a finite generating set. Iterating the period identities, including their inverses, extends them to the transverse subgroup. The relative semantic fields are already invariant under $K_{\text{GT}}^{(\ell)}$ by (5.5); this proves (i)–(ii). P4 gives $\partial_h S = 0$ for all structural role generators. Taking ∂_t and commuting differences gives $\partial_h \zeta = 0$, proving (iii).

For (iv), P1–P2 include the field direction t , hence both primary tracks appearing in $U_j^\pm$ have period t . P4 gives the required e_ℓ -period to the S coordinate and therefore to $(\lambda_j \circ \text{pr}_S)\alpha$. Finally P3 gives the two displayed periods to the raw coordinate $V_{n_1, n_2}\alpha = \lambda_{n_1, n_2}(S)$. Commuting each with ∂_t gives the same periods to $\lambda_{n_1, n_2}(\zeta)$; iteration proves (v). $\square$

We next identify each structural Boolean field with the corresponding coordinate of the finite-field state. Fix $j = (\ell, a, b, n) \in \mathcal{I}_{\text{sem}}$, and let e_ℓ be the common Boolean flip in the ℓ th copy. By Claim 5.4(i), the raw primary tracks of this copy factor through the projection onto $G_{\text{GT}}^{(\ell)}$. Their restriction to that factor satisfies S_{ref} . Choose a compatible Boolean normalization of this tuple, denote its common odd unit and normalized tracks by $v_\ell, \kappa_*^\ell, (\kappa_{a,b,n}^\ell)_{(a,b,n) \in \mathcal{A}_0}$, and extend the normalized tracks constantly over the transverse subgroup. Thus $\tilde{\xi}_j = \kappa_{a,b,n}^\ell \oplus \kappa_*^\ell$. Compatible Booleanity and (5.3) give, pointwise,

$$\partial_{e_\ell}(U_j^- \alpha) = 2v_\ell(\kappa_*^\ell - \kappa_{a,b,n}^\ell), \quad \text{and} \quad \partial_{e_\ell}(U_j^+ \alpha) = 2v_\ell(1 - \kappa_{a,b,n}^\ell - \kappa_*^\ell).$$

Since v_ℓ is odd and $M_{\text{str}} = 26$, the only possible nonzero values on the right are $\pm 2v_\ell$, which are nonzero in $C_{2M_{\text{str}}}$. Therefore $\partial_{e_\ell}(U_j^- \alpha) = 0 \iff \tilde{\xi}_j = 0$, and $\partial_{e_\ell}(U_j^+ \alpha) = 0 \iff \tilde{\xi}_j = 1$. Define $V_j := \lambda_j \circ \text{pr}_S$ and $\beta_j := \partial_t(V_j \alpha) = \lambda_j(\zeta)$. The maps (U_j^-, V_j) and (U_j^+, V_j) are jointly surjective. Both U_j^- and U_j^+ are surjective because the corresponding primary and reference coordinate projections are independent direct-product coordinates, while V_j is surjective on the independent field factor of $H_{\mathcal{W}}$.

By Claim 5.4(iv), the cross-period hypotheses of Lemma 5.3 hold with $r = e_\ell$ and $s = t$. Append the two rectangle rows expressing

$$\partial_{e_\ell}(U_j^- \alpha) = 0 \quad \text{or} \quad \beta_j = 1, \quad \text{and} \quad \partial_{e_\ell}(U_j^+ \alpha) = 0 \quad \text{or} \quad \beta_j = 0. \quad (5.6)$$

Together with the two detector equivalences above, these rows are equivalent to $\beta_j = \tilde{\xi}_j$ on G_{str} . Indeed, when $\tilde{\xi}_j = 0$, the second row forces $\beta_j = 0$; when $\tilde{\xi}_j = 1$, the first row forces $\beta_j = 1$. The other row is automatic in each case.

We can now state and prove the bounded-generator compiler.

Theorem 5.5. *There is a total computable map sending every finite Wang instance $\mathcal{W}$ to a finite fiberwise partition system $\mathcal{E}_{\mathcal{W}}$ for one function $\alpha: G_{\mathcal{W}} \rightarrow H_{\mathcal{W}}$, where $G_{\mathcal{W}} = \mathbb{Z}^4 \times K_{\mathcal{W}}$, such that $\text{gen}(K_{\mathcal{W}}) \leq m_0 + 7$ and $\text{gen}(H_{\mathcal{W}}) \leq \text{gen}(H_{\text{str}}) + 2m_0$, and such that $\mathcal{E}_{\mathcal{W}}$ is solvable if and only if $\mathcal{W}$ tiles $\mathbb{Z}^2$. For nonempty $\mathcal{W}$ one may take $K_{\mathcal{W}} = K_{\text{str}} \times \mathbb{K}_{\mathcal{W}}^+ \times C_{p_{\mathcal{W}}}$ and $H_{\mathcal{W}} = H_{\text{str}} \times \mathbb{K}_{\mathcal{W}}^2$. The same rank bounds are used in the empty-input branch.*

Proof. We first describe the algorithm. If the Wang alphabet $\mathcal{W}$ is empty, take $p_{\mathcal{W}} = 3$, construct $\mathbb{K}_{\mathcal{W}} = \mathbb{F}_{3^{m_0}}$, use the groups specified in the theorem, and output the impossible row $(\alpha(x) + H_{\mathcal{W}}) \uplus (\alpha(x) + H_{\mathcal{W}}) = H_{\mathcal{W}}$. This row has no solution, exactly as an empty Wang alphabet has no tiling.

Suppose now that $\mathcal{W}$ is nonempty. Compute R_1, R_2 , the prime $p_{\mathcal{W}}$, the field $\mathbb{K}_{\mathcal{W}}$, and the complete finite relation $\Theta_{\mathcal{W}}$. Output the following four families of rows.

- (A1) the two fixed structural systems, with their shifts embedded in $G_{\mathcal{W}}$ and their value groups cylinder-lifted to $H_{\mathcal{W}}$;
- (A2) all field-rack rows and the forbidden-state rows specified above;
- (A3) all pin rows P1–P4;
- (A4) the two bridge rows in (5.6) for every $j \in \mathcal{I}_{\text{sem}}$.

Every subgroup, complement, quotient cylinder, and translated block occurring in these four families is output as a complete list. The prime is found by trial division; the field construction terminates by Lemma 4.1; and $\Theta_{\mathcal{W}}$ is computed by finite enumeration. Thus this construction is a total algorithm in the encoding of Definition 2.1.

We prove soundness for an arbitrary solution α ; no periodicity or computability of the solution is assumed. On every coset of each embedded copy $G_{\text{GT}}^{(\ell)}$, the corresponding projection solves its fixed system. By P1–P2, the primary data of copy ℓ depend only on its own $\mathbb{Z}^2 \times K_{\text{GT}}$ coordinates. Hence Proposition 5.1(iii) decodes two uniquely determined good-column Sudoku boards Ξ^1, Ξ^2 , and Proposition 5.1(iv) identifies their semantic fields with the $\tilde{\xi}_j$.

The field rack gives a state ζ , and P4 makes it independent of all structural role coordinates. The forbidden-state rows and Theorem 4.4 give $\text{coord}_{\mathcal{W}}(\zeta(x)) \in \Theta_{\mathcal{W}}$ at every $x \in \mathbb{Z}^4$. For every structural index, the bridge rows and the detector equivalences give $\beta_j = \tilde{\xi}_j$; at role zero this is $\beta_j = \xi_j$.

Write a point of $\mathbb{Z}^4$ as $x = (x_1, x_2)$, where $x_{\ell} = (i_{\ell}, j_{\ell}) \in \mathbb{Z}_{(\ell)}^2$. For $(n_1, n_2) \in [N]^2$, let $\omega_{n_1, n_2}(x) := \lambda_{n_1, n_2}(\zeta(x))$. Condition (b) in $\Theta_{\mathcal{W}}$ makes this a valid tile code. P3 shows that it depends only on $m_1 := j_1 n_1 + i_1$ and $m_2 := j_2 n_2 + i_2$. Indeed, two points $x, x' \in \mathbb{Z}^4$ giving the same values of m_1 and m_2 differ by integer multiples of $-n_1 v_{1,1} + v_{1,2}$ and $-n_2 v_{2,1} + v_{2,2}$. We may therefore define a single function $\Upsilon: \mathbb{B} \times \mathbb{B} \rightarrow \mathcal{W}$ by $\Upsilon((n_1, m_1), (n_2, m_2)) := \text{dec}_{\mathcal{W}}(\omega_{n_1, n_2}(x))$.

Fix a second-board cell (n_2, m_2) and a Sudoku line of the first board, written $m_1 = j_1 n_1 + i_1$. Choose any i_2, j_2 with $m_2 = j_2 n_2 + i_2$ and evaluate the condition $\text{coord}_{\mathcal{W}}(\zeta(x)) \in \Theta_{\mathcal{W}}$ at $x = ((i_1, j_1), (i_2, j_2))$. The bridge equality identifies the structural word μ_b^1 in condition (c) with $n_1 \mapsto \Xi^1(n_1, j_1 n_1 + i_1)$, while the definition of Υ identifies its decoration with $n_1 \mapsto \Upsilon((n_1, j_1 n_1 + i_1), (n_2, m_2))$. Hence that line lies in $\mathcal{D}_{R_1}$. Since the choices were arbitrary, every required line of the first board is correctly decorated. The same argument with the two factors reversed and condition (d) gives every required R_2 -decorated line of the second board. Thus (Ξ^1, Ξ^2, Υ) is a bi-decorated solution. By Proposition 3.2, $\mathcal{W}$ tiles $\mathbb{Z}^2$.

Conversely, suppose that $\mathcal{W}$ tiles $\mathbb{Z}^2$. The implication from a Wang tiling to a bi-decorated solution in Proposition 3.2 gives two standard good-column boards and a common decoration Υ . For each board, use Proposition 5.1(iii) to choose a primary-and-witness solution of the fixed phase-pinned structural system that decodes to it. Extend the complete tuple of each copy constantly over the other structural copy and $\mathcal{R}_{\mathcal{W}}$. This satisfies the structural equations and P1–P2.

On $\mathbb{Z}^4$, identified with the role-zero subgroup, define the state vector $\vartheta^{\circ}(x) \in \mathbb{F}_{p_{\mathcal{W}}}^{\mathcal{I}_0}$ as follows. For $j = (\ell, a, b, n) \in \mathcal{I}_{\text{sem}}$, set the j th coordinate of $\vartheta^{\circ}(x)$ equal to $\xi_j(x_{\ell})$. For $(n_1, n_2) \in \mathcal{I}_{\text{dec}}$, set its (n_1, n_2) th coordinate equal to

$$\text{code}_{\mathcal{W}}(\Upsilon((n_1, j_1 n_1 + i_1), (n_2, j_2 n_2 + i_2))).$$

By construction $\vartheta^{\circ}(x) \in \Theta_{\mathcal{W}}$. Put $\zeta^{\circ}(x) := \text{coord}_{\mathcal{W}}^{-1}(\vartheta^{\circ}(x))$ and extend ζ° constantly over K_{str} . Choose the unrestricted values of $P|_{G_{\text{str}}}$ and $S|_{G_{\text{str}}}$ in Lemma 4.2 to be zero. Then, for $g \in G_{\text{str}}$, $u \in \mathbb{K}_{\mathcal{W}}^+$, and $\tau \in \mathbb{F}_{p_{\mathcal{W}}}$, set $S(g + u + \tau t) = \tau \zeta^{\circ}(g)$ and $P(g + u + \tau t) = u \zeta^{\circ}(g)$. These functions satisfy all field-rack rows by Lemma 4.2. Since every state lies in $\Theta_{\mathcal{W}}$, all forbidden-state rows hold. Each decoration coordinate of $\vartheta^{\circ}(x)$ depends on x only through $(j_1 n_1 + i_1, j_2 n_2 + i_2)$, so P3 holds; the extension over structural roles gives P4. Finally, the definition of ϑ° , the constant extension of ζ° over K_{str} , and the K_{str} -invariance of ξ_j give $\beta_j = \lambda_j(\zeta^{\circ}) = \tilde{\xi}_j$ on G_{str} . Hence each pair of bridge disjunctions is true, and Lemma 5.3 gives the bridge rows. Thus $\mathcal{E}_{\mathcal{W}}$ is solvable.

The additive field group has rank m_0 , and the extra cyclic role has rank one. Since $K_{\text{str}} = (C_2)^6$,

$$\text{gen}(K_{\mathcal{W}}) \leq 6 + m_0 + 1, \quad \text{gen}(H_{\mathcal{W}}) \leq \text{gen}(H_{\text{str}}) + 2m_0. \quad (5.7)$$

The estimates (5.7) prove the asserted rank bounds; together with the soundness and completeness arguments, they complete the proof. $\square$

6 From fiberwise partitions to fixed-dimensional undecidability

This section converts the fiberwise partition system into one tile. We first simplify its equations, then convert solutions into common tiling complements via their graphs and apply rigid stacking over a finitely generated abelian group of the form $\mathbb{Z}^4 \times K$, and finally compute coset representatives and apply the theorem of Meyerovitch, Sanadhya, and Solomon [17, Theorem 1.1] to obtain an equivalent tile in $\mathbb{Z}^d$. None of these steps assumes that a solution is periodic. Every intermediate tiling equation has the whole group as its target; no periodic target subset is introduced.

Lemma 6.1. *Let $\mathcal{E}$ be a finite fiberwise partition system on $\alpha: G \rightarrow H$. There is a total algorithm that produces exactly one of the following outcomes.*

- (i) *It finds a row that no α can satisfy, thereby proving that $\mathcal{E}$ has no solution.*
- (ii) *It writes every row in the equivalent form $\biguplus_{j=1}^{J_i} (\alpha(x + h_{i,j}) + E_{i,j}) = H$ for every $x \in G$ and $i \in \{1, \dots, M\}$, where $J_i \geq 1$, every $E_{i,j}$ is nonempty, and, for each fixed i , the shifts $h_{i,1}, \dots, h_{i,J_i}$ are pairwise distinct. In this case, put $F_0 := \{0\} \times H$ and, for $1 \leq i \leq M$, define $F_i := \biguplus_{j=1}^{J_i} (\{-h_{i,j}\} \times E_{i,j}) \subseteq G \times H$. Then $\mathcal{E}$ has a solution if and only if there exists a set $A \subseteq G \times H$ satisfying $A \oplus F_i = G \times H$ for every $i \in \{0, \dots, M\}$. Every such A is the graph of a solution to $\mathcal{E}$.*

Proof. For each row, delete every empty block and partition the remaining summands according to their actual shift in G . Equality of shifts, including torsion coordinates, is decidable from Definition 2.1 by Smith normal form. Within one shift class, two overlapping blocks contribute the same translate of their intersection twice for every x , so the row cannot be satisfied by any α . If the blocks in the class are pairwise disjoint, replace them by their union; this preserves exact coverage. If no nonempty block remains, the left side is empty while H is nonempty, so the row again cannot be satisfied by any α . Otherwise these operations leave nonempty blocks at pairwise distinct shifts and preserve the row. Finding such an unsatisfiable row gives outcome (i); otherwise these operations give outcome (ii).

Suppose outcome (ii) occurs. The proof of Greenfeld–Tao [9, Theorem 4.1], with its aperiodicity conclusion omitted, gives the required equivalence. If A is a common complement, the equation for F_0 forces each vertical fiber $\{x\} \times H$ to meet A once, so $A = \{(x, \alpha(x)) : x \in G\}$ for a unique map α . Restricting the equation for F_i to the vertical fiber above x , for $1 \leq i \leq M$, gives exactly row i of $\mathcal{E}$, so α solves $\mathcal{E}$. Conversely, if α satisfies $\mathcal{E}$, the same fiberwise calculation shows that its graph is a common tiling complement for $F_0, \dots, F_M$. $\square$

Lemma 6.2. *Given finite nonempty $F_0, \dots, F_M \subseteq \Gamma$, one can effectively find a power of two L and a partition $C_L = P_0 \uplus \dots \uplus P_M$ into nonempty sets such that the one tile $\tilde{F} := \biguplus_{i=0}^M (F_i \times P_i)$ tiles $\Gamma \times C_L$ if and only if the F_i have a common tiling complement in Γ .*

Proof. Greenfeld–Tao supply a partition with the following property; see [9, Lemma 3.2]. For all $0 \leq i, j \leq M$ and every $t \in C_L \setminus \{0\}$,

$$P_i \cap (P_j + t) \neq \emptyset. \quad (6.1)$$

Their proof gives more than preservation of aperiodicity; it proves the existence equivalence stated here; see [9, Theorem 3.1]. A common complement A gives the tiling $(A \times \{0\}) \oplus \tilde{F} = \Gamma \times C_L$. Conversely, let $B \subseteq \Gamma \times C_L$ satisfy $B \oplus \tilde{F} = \Gamma \times C_L$. For $g \in \Gamma$ and $0 \leq i \leq M$, define $\mathcal{R}_{g,i} := \{((a, t), f) \in B \times F_i : a + f = g\}$. If two distinct elements of $\mathcal{R}_{g,i}$ have different C_L -coordinates $t \neq t'$, then $t + P_i$ and $t' + P_i$ intersect by (6.1). If their C_L -coordinates are equal, the same nonempty translate $t + P_i$ occurs twice. Both alternatives contradict exactness. Hence $|\mathcal{R}_{g,i}| \leq 1$. Restricting the tiling equation to the vertical fiber $\{g\} \times C_L$ gives the exact partition

$$C_L = \biguplus_{i=0}^M \biguplus_{((a,t),f) \in \mathcal{R}_{g,i}} (t + P_i).$$

Counting this fiber gives

$$L = \sum_{i=0}^M |\mathcal{R}_{g,i}| |P_i| \leq \sum_{i=0}^M |P_i| = L,$$

and, because every P_i is nonempty, equality forces $|\mathcal{R}_{g,i}| = 1$ for every g, i .

We next show that the projection $B \rightarrow \Gamma$ is injective. Suppose that $(a, t), (a, t') \in B$ are distinct. Then $t \neq t'$. Choose any i and any $f \in F_i$, which is possible because every F_i is nonempty. The two copies of the tile contribute the intersecting sets $\{a + f\} \times (t + P_i)$ and $\{a + f\} \times (t' + P_i)$, contradicting exactness. Thus $A := \text{pr}_\Gamma(B)$ is in bijection with B . The identity $|\mathcal{R}_{g,i}| = 1$ now says precisely that every g has a unique representation $g = a + f$ with $a \in A$ and $f \in F_i$. Hence $A \oplus F_i = \Gamma$ for every i .

Effectivity does not rely on the probabilistic proof. Enumerate powers $L = 2^r$ and all labeled partitions of C_L into $M + 1$ nonempty sets, and test (6.1). This enumeration eventually finds a suitable partition; see [9, Lemma 3.2]. $\square$

Apply the algorithm in Lemma 6.1 to $\mathcal{E}_W$. If it returns a common-complement system, retain that system. If it instead proves that $\mathcal{E}_W$ is inconsistent, use the following inconsistent common-complement system in $\Gamma_0 := G_W \times H_W$ given by $A \oplus \{0\} = \Gamma_0$ and $A \oplus \{0, e_1\} = \Gamma_0$, where e_1 is the first free basis vector. The first equation forces $A = \Gamma_0$, which makes the second equation non-exact. In both cases, the resulting common-complement family has a common tiling complement exactly when $\mathcal{E}_W$ has a solution. Both families use the same group and hence the same generator bound. Applying Lemma 6.2 to the resulting family, we obtain one finite tile $\tilde{F}_W \subseteq \Gamma_W$, where $\Gamma_W := \mathbb{Z}^4 \times K_W \times H_W \times C_{L_W}$, such that

$$\tilde{F}_W \text{ tiles } \Gamma_W \iff \mathcal{E}_W \text{ is solvable.} \quad (6.2)$$

Here $K_{\text{str}} = (\mathbb{Z}/2\mathbb{Z})^6$, while H_{str} is the product of the two fixed structural value groups chosen in Proposition 5.1. The factor $\mathbb{K}_W^+$ contributes m_0 generators, $C_{p_W}t$ contributes one, $\mathbb{K}_W^2$ contributes $2m_0$, and C_{L_W} contributes one. Accordingly, set

$$d_* := 4 + 6 + (m_0 + 1) + \text{gen}(H_{\text{str}}) + 2m_0 + 1 = 12 + \text{gen}(H_{\text{str}}) + 3m_0. \quad (6.3)$$

For every finite abelian group L_0 and every integer $r \geq 0$, $\text{gen}(\mathbb{Z}^r \times L_0) \leq r + \text{gen}(L_0)$. Every quantity on the right-hand side of (6.3) is fixed before the Wang instance is given. Once the structural fiberwise partition systems and their finite value groups are chosen and encoded, an invariant-factor presentation of H_{str} determines $\text{gen}(H_{\text{str}})$ and hence a numerical value of d_* .

The prime p_W , the order of $\mathbb{K}_W$, and the stacking modulus L_W may grow with the input. They affect group orders but not the fixed generator ranks m_0 and $2m_0$ of $\mathbb{K}_W^+$ and $\mathbb{K}_W^2$, respectively; the remaining factors $C_{p_W}t$ and C_{L_W} are cyclic. The generator-rank bounds (5.7) in Theorem 5.5 give $\text{gen}(\Gamma_W) \leq 4 + \text{gen}(K_W) + \text{gen}(H_W) + 1 \leq d_*$.

Choose $f_0 \in \tilde{F}_W$ and replace the tile by $F_W^0 := \tilde{F}_W - f_0$. Then $0 \in F_W^0$, and $A \oplus \tilde{F}_W = \Gamma_W$ if and only if $(A + f_0) \oplus F_W^0 = \Gamma_W$. Using the direct-product descriptions of K_W , H_W , and C_{L_W} , compute an explicit generating list for Γ_W of length at most d_* . Append copies of the zero element until the list has length exactly d_* , and map the standard basis to this list. This gives an effective surjection

$$\rho_W: \mathbb{Z}^{d_*} \rightarrow \Gamma_W. \quad (6.4)$$

We now spell out the required coset-representative algorithm, including the case where the kernel has rank smaller than d_* .

Lemma 6.3. *Let Γ be a finitely generated abelian group given by a finite integer presentation, and let $\rho: \mathbb{Z}^d \rightarrow \Gamma$ be specified by the images of the standard basis. One can compute the following data.*

- (i) a $\mathbb{Z}$ -basis of $L = \ker \rho$;
- (ii) a decidable transversal $D \subseteq \mathbb{Z}^d$ meeting every coset of L in exactly one point, with $0 \in D$;
- (iii) an algorithm that takes $\gamma \in \Gamma$ and returns the unique $\sigma(\gamma) \in D$ satisfying $\rho(\sigma(\gamma)) = \gamma$.

The transversal D is generally infinite when L is not of full rank.

Proof. Fix deterministic Smith- and Hermite-normal-form algorithms throughout this proof; the set-theoretic section constructed below is relative to those fixed choices. Write the presentation as $\Gamma = \mathbb{Z}^m / Q\mathbb{Z}^s$, where Q is an integer $m \times s$ matrix, and let A be the $m \times d$ matrix representing ρ . The kernel of the integer matrix $[A \ -Q]: \mathbb{Z}^{d+s} \rightarrow \mathbb{Z}^m$ is computable by Smith normal form. Project a computed set of generators for this kernel to its first d coordinates, form the integer matrix whose columns are the projected vectors, and compute its column Hermite normal form. After deleting the zero columns, the remaining columns form a $\mathbb{Z}$ -basis matrix $B_L = (k_1, \dots, k_\ell)$ for L .

Apply Smith normal form to the injective matrix $B_L: \mathbb{Z}^\ell \rightarrow \mathbb{Z}^d$. Thus one computes unimodular matrices $U \in \text{GL}_d(\mathbb{Z})$ and $V \in \text{GL}_\ell(\mathbb{Z})$ and positive integers $s_1 \mid \dots \mid s_\ell$ such that

$$UB_LV = \begin{pmatrix} \text{diag}(s_1, \dots, s_\ell) \\ 0_{(d-\ell) \times \ell} \end{pmatrix}.$$

In the U -coordinates, L is generated by $s_r e_r$ for $1 \leq r \leq \ell$. Consequently

$$D := U^{-1} \{y \in \mathbb{Z}^d : 0 \leq y_r < s_r \ (1 \leq r \leq \ell)\}$$

is a transversal; the coordinates $y_{\ell+1}, \dots, y_d$ are unrestricted. Reducing the first ℓ coordinates of Ux modulo s_r gives the unique member of D in $x + L$.

Finally represent γ by a vector $g \in \mathbb{Z}^m$. Surjectivity of ρ ensures that the integer system $Ax - Qz = g$ has a solution. Smith normal form of $[A \ -Q]$ computes one such $x \in \mathbb{Z}^d$; reducing x by the preceding rule gives $\sigma(\gamma)$. Changing the representative g or the chosen solution x changes x by an element of L , so the reduced representative is unchanged. The zero vector reduces to zero. $\square$

Apply [Lemma 6.3](#) to (6.4). It computes a $\mathbb{Z}$ -basis of $\ker \rho_W$, a finite Smith-normal-form description of a possibly infinite transversal D , and the coset-representative map $\gamma \mapsto \sigma(\gamma)$. For an abelian group Λ and a finite tile $F \subseteq \Lambda$ containing zero, write $\text{Tile}_0(F; \Lambda) := \{A \subseteq \Lambda : 0 \in A, F \oplus A = \Lambda\}$. The following theorem supplies the final step of the reduction.

Theorem 6.4 ([17], Theorem 1.1). *Let $d \geq 2$, let Γ be a finitely generated abelian group, let $\rho: \mathbb{Z}^d \twoheadrightarrow \Gamma$, and let D be a transversal (fundamental domain) for $\ker \rho$ with $0 \in D$. For every finite $F \subseteq \Gamma$ with $0 \in F$, there exist an integer $N_{\text{MSS}} \geq 1$ and a finite $\hat{F} \subseteq \mathbb{Z}^d$ with $0 \in \hat{F}$ such that*

$$\text{Tile}_0(\hat{F}; \mathbb{Z}^d) = \{N_{\text{MSS}}\rho^{-1}(A) : A \in \text{Tile}_0(F; \Gamma)\}. \quad (6.5)$$

Here $\rho^{-1}(A)$ is the full set-theoretic preimage and $N_{\text{MSS}}\rho^{-1}(A) := \{N_{\text{MSS}}x : x \in \rho^{-1}(A)\}$ is coordinate dilation. In particular, F tiles Γ if and only if $\hat{F}$ tiles $\mathbb{Z}^d$.

Proof. Equation (6.5) is the displayed equality in the case $k = 1$ of the cited theorem, specialized to one tile. In the notation used there, specialize their lattice parameter to $\ker \rho$, and let $T, T_1, T_2 \subseteq \mathbb{Z}^d$ be the rigid tiles supplied by [17, Lemma 2.1] with $s = 2$; the scale in that lemma is N_{MSS} . Also put $R_F := \rho^{-1}(F \setminus \{0\}) \cap D$. For $k = 1$ their tile is exactly

$$\hat{F} = (N_{\text{MSS}}R_F \oplus T_1) \uplus T_2. \quad (6.6)$$

Moreover, $0 \in T_2$, and hence $0 \in \hat{F}$; see [17, Lemma 2.1].

Normalization causes no loss. If $F \oplus A = \Gamma$, choose $a_0 \in A$. Then $F \oplus (A - a_0) = \Gamma$ and $0 \in A - a_0$. The same argument applies in $\mathbb{Z}^d$, so the stated tiling equivalence follows from (6.5). $\square$

Lemma 6.5. *Assume the hypotheses of Theorem 6.4, except that D is not prescribed. Suppose that Γ is given by a finite integer presentation, ρ is specified by the images of the standard basis of $\mathbb{Z}^d$, and the elements of F are given by integer representatives. Then one can compute the following data:*

- (i) the integer N_{MSS} ;
- (ii) the Smith-normal-form parameters $U, s_1, \dots, s_\ell$ defining a transversal D ;
- (iii) a membership test for D ;
- (iv) the representative map $\sigma: \Gamma \rightarrow D$;
- (v) the complete coordinate list of $\hat{F}$.

Proof. First apply [Lemma 6.3](#). This computes a lattice basis of $\ker \rho$, the matrix U and integers $s_1, \dots, s_\ell$ defining the transversal D , and the representative map. Hence the finite set $R_F = \{\sigma(f) : f \in F \setminus \{0\}\}$ is computable.

We now make the construction in [\[17, Lemma 2.1\]](#) explicit in the case $s = 2$. Put $r := \text{rank}(\ker \rho)$, and let $w_1, \dots, w_r$ be the computed basis of $\ker \rho$. For $n \geq 0$, set $Q_n := \{-n, \dots, n\}^d$, and, for $n \geq 1$, set $S_n := Q_n \setminus Q_{n-1}$. Enumerate integers $m \geq 1$ in increasing order and, for each m , enumerate the finite set Q_m^{d+2r} of tuples $(v_1, \dots, v_{d+2r})$ until $v_i + Q_{d+2r} \subseteq Q_m$ for every $1 \leq i \leq d+2r$ and $(v_i + Q_{d+2r}) \cap (v_j + Q_{d+2r}) = \emptyset$ for every $1 \leq i < j \leq d+2r$. These are finite conditions decidable by integer arithmetic. The existence argument in [\[17, Lemma 2.1\]](#) guarantees that the search terminates.

For the first tuple found, set $N_{\text{MSS}} := 2m + 1$ and define

$$T := \left(Q_m \setminus \biguplus_{i=1}^d (v_i + S_i) \right) \uplus \biguplus_{i=1}^d (N_{\text{MSS}} e_i + v_i + S_i),$$

where $e_1, \dots, e_d$ are the standard basis vectors. For $j = 1, 2$, define

$$T_j := \left(T \setminus \biguplus_{h=1}^r (v_{d+r(j-1)+h} + S_{d+r(j-1)+h}) \right) \uplus \biguplus_{h=1}^r (N_{\text{MSS}} w_h + v_{d+r(j-1)+h} + S_{d+r(j-1)+h}).$$

These are exactly the finite sets in formulas (3)–(4) of [\[17, Lemma 2.1\]](#), so their complete coordinate lists are computable. Substituting T_1, T_2 into (6.6) gives every coordinate of $\widehat{F}$. Thus no periodicity assumption or nonconstructive choice is used; every choice is made by a finite search or an explicit integer calculation. This is also the construction used by Meyerovitch, Sanadhya, and Solomon [\[17, Corollary 1.3\]](#) in their algorithmic reduction. $\square$

The finite coordinate list of $F_{\mathcal{W}}^0$, the presentation of $\Gamma_{\mathcal{W}}$, and the quotient map (6.4) satisfy the effective hypotheses above; in particular, $d = d_* \geq 12 \geq 2$. [Lemma 6.3](#) computes the Smith-normal-form description of D and the representative $\sigma(\gamma)$ for any γ , [Theorem 6.4](#) supplies the equality of normalized tiling-complement sets, and [Lemma 6.5](#) computes N_{MSS} and the coordinate list of the tile $\widehat{F}_{\mathcal{W}} \subseteq \mathbb{Z}^{d_*}$ in [Theorem 1.1](#).

Proof of Theorem 1.1. The construction in [Theorem 5.5](#) produces a solvable fiberwise partition system exactly when $\mathcal{W}$ tiles $\mathbb{Z}^2$. If graph conversion returns a common-complement system, [Lemmas 6.1](#) and [6.2](#) give (6.2). If graph conversion instead proves inconsistency, the inconsistent fallback system gives the same equivalence. [Theorem 6.4](#) and [Lemma 6.5](#) then show that $F_{\mathcal{W}}^0$ tiles $\Gamma_{\mathcal{W}}$ if and only if $\widehat{F}_{\mathcal{W}}$ tiles $\mathbb{Z}^{d_*}$. Their composition is therefore a total computable many-one reduction from Wang tileability to TILE_{d_*} . If TILE_{d_*} were decidable, applying a decision algorithm to $\widehat{F}_{\mathcal{W}}$ would decide the domino problem for Wang tiles, contrary to Berger’s theorem [\[2\]](#).

To pass to larger dimensions, for a finite $F \subseteq \mathbb{Z}^d$ put $F^\uparrow := F \times \{0\} \subseteq \mathbb{Z}^{d+1}$. If $F \oplus A = \mathbb{Z}^d$, then $F^\uparrow \oplus (A \times \mathbb{Z}) = \mathbb{Z}^{d+1}$. Conversely, suppose $F^\uparrow \oplus A' = \mathbb{Z}^{d+1}$. For each $t \in \mathbb{Z}$, put $A'_t := \{x \in \mathbb{Z}^d : (x, t) \in A'\}$. Since every element of $F^\uparrow$ has last coordinate zero, A'_t is a tiling complement for F . Hence F tiles $\mathbb{Z}^d$ if and only if $F^\uparrow$ tiles $\mathbb{Z}^{d+1}$. For each fixed $d \geq d_*$, iterating this computable lifting $d - d_*$ times gives a total computable many-one reduction from TILE_{d_*} to TILE_d . Therefore TILE_d is undecidable for every fixed $d \geq d_*$. $\square$

7 Concluding remarks

We now bound the generator rank of the witness-unrolled extension fixed in the proof of [Proposition 5.1](#) and record an explicit but deliberately coarse dimension bound. All structural tracks counted below belong to the fixed Greenfeld–Tao structural library and are independent of the Wang instance. The input-dependent state relation $\Theta_{\mathcal{W}}$ is compiled separately by the finite-field construction of [Section 4](#). In the fixed extension, the two-track construction of [\[9, proof of Proposition 6.5\]](#), applied to a compatible Boolean constraint involving L raw tracks, introduces at most $6(L - 1)$ auxiliary $C_{2M_{\text{str}}}$ -valued functions. The contributions to $\text{gen}(H_{\text{GT}})$ are as follows.

- (i) The primary functions contribute R_0 .

- (ii) For each orbit under simultaneous complementation that is disjoint from $\tilde{\Omega}$, the selected construction in [9, proof of Proposition 6.7] introduces $R_0 - 2$ further tracks and imposes compatibility on all $2R_0 - 2$ tracks. It therefore uses at most $(R_0 - 2) + 6(2R_0 - 3) = 13R_0 - 20$ auxiliary functions per omitted orbit. There are at most 2^{R_0-1} such orbits. Since R_0 is odd, the preliminary padding in that proof is unnecessary.
- (iii) Each of the N raw-track Boolean periodized-permutation constraints in [9, Proposition 6.9] contributes at most $6(s_0 - 1) = 54$ auxiliary functions.
- (iv) The period constraints and reference pins contribute no additional value coordinates.

Every primary or auxiliary function counted above is $C_{2^{M_{\text{str}}}}$ -valued, so its value-group factor is cyclic and contributes at most one generator to H_{GT} . Hence

$$\text{gen}(H_{\text{GT}}) \leq R_0 + (13R_0 - 20)2^{R_0-1} + 54N.$$

Since $H_{\text{str}} = H_{\text{GT}}^{(1)} \times H_{\text{GT}}^{(2)}$, subadditivity of gen , together with $R_0 = 20\,971\,521$, $N = 1\,048\,576$, and $m_0 = 1\,099\,553\,570\,816$, gives through (6.3)

$$d_* \leq 272\,629\,753 \cdot 2^{20\,971\,521} + 3\,298\,815\,901\,710 < 2^{20\,971\,550}.$$

Thus $2^{20\,971\,550}$ is a valid explicit threshold in Theorem 1.1. This count concerns the witness-unrolled expressible extension fixed in the proof of Proposition 5.1, not an arbitrary expressible extension to which redundant finite factors have been adjoined.

Finally, Proposition 7.1 locates the decision problem in the arithmetical hierarchy.

Proposition 7.1. *For every fixed d , the language TILE_d belongs to Π_1^0 .*

Proof. Equivalently, non-tileability is computably enumerable: whenever a finite set does not tile $\mathbb{Z}^d$, a finite obstruction can eventually be found by exhaustive search. To verify the classification, let $X_d := \{0, 1\}^{\mathbb{Z}^d}$ with the product topology. For a finite $F \subseteq \mathbb{Z}^d$ and $n \in \mathbb{N}$, put $B_n := [-n, n]^d \cap \mathbb{Z}^d$ and define

$$\mathcal{C}_n(F) := \left\{ \theta \in X_d : \sum_{f \in F} \theta(x - f) = 1 \text{ for every } x \in B_n \right\}.$$

Membership in $\mathcal{C}_n(F)$ depends only on the finite coordinate set $B_n - F := \{x - f : x \in B_n, f \in F\}$, so $\mathcal{C}_n(F)$ is clopen. Let $S(F, n)$ assert that there is a map $\theta : B_n - F \rightarrow \{0, 1\}$ satisfying $\sum_{f \in F} \theta(x - f) = 1$ for every $x \in B_n$. The predicate $S(F, n)$ is decidable uniformly from the encodings of F and n by finite enumeration, and $S(F, n)$ holds exactly when $\mathcal{C}_n(F)$ is nonempty. Moreover, $\mathcal{C}_{n+1}(F) \subseteq \mathcal{C}_n(F)$. If $S(F, n)$ holds for every $n \in \mathbb{N}$, compactness of X_d gives a point in $\bigcap_{n \in \mathbb{N}} \mathcal{C}_n(F)$, which is the indicator of a global exact tiling complement. Consequently, $F \in \text{TILE}_d$ if and only if $S(F, n)$ holds for every $n \in \mathbb{N}$. $\square$

The same reduction gives a concrete independence consequence. Fix a Gödel coding of ZFC and let $\mathcal{R}_{\text{ZFC}}$ be a Rosser sentence for this coding. Since $\mathcal{R}_{\text{ZFC}}$ is a Π_1^0 sentence, a Turing machine can search for a counterexample and halt exactly when $\mathcal{R}_{\text{ZFC}}$ is false. Composing Berger's machine simulation, Theorem 1.1, and the dimension-lifting construction in its proof gives, for every fixed $d \geq d_*$, an effectively specified finite set $F_{\text{ZFC}, d} \subseteq \mathbb{Z}^d$ such that ZFC proves that $F_{\text{ZFC}, d}$ tiles $\mathbb{Z}^d$ if and only if $\mathcal{R}_{\text{ZFC}}$. If ZFC is consistent, Rosser's incompleteness theorem [21] implies that ZFC proves neither this tiling statement nor its negation. In the standard model of arithmetic, $\mathcal{R}_{\text{ZFC}}$ is then true, so $F_{\text{ZFC}, d}$ does tile $\mathbb{Z}^d$. Thus, subject only to the consistency of ZFC, every fixed dimension $d \geq d_*$ admits a finite monotile whose tileability is independent of ZFC.

Declaration on the use of AI

The authors used generative AI tools to discuss proof strategies, check proofs, and improve the exposition.

References

- [1] S. O. Aanderaa and H. R. Lewis. Linear sampling and the $\forall\exists\forall$ case of the decision problem. *J. Symbolic Logic*, 39(3):519–548, 1974.
- [2] R. Berger. *The undecidability of the domino problem*, volume 66 of *Mem. Amer. Math. Soc.* American Mathematical Society, Providence, RI, 1966.
- [3] S. Bhattacharya. Periodicity and decidability of tilings of $\mathbb{Z}^2$. *Amer. J. Math.*, 142(1):255–266, 2020.
- [4] T. Coulbois, A. Gajardo, P. Guillon, and V. H. Lutfalla. Aperiodic monotiles: from geometry to groups. *Theoret. Comput. Sci.*, 1059:Paper No. 115555, 17, 2026.
- [5] E. M. Coven and A. D. Meyerowitz. Tiling the integers with translates of one finite set. *J. Algebra*, 212(1):161–174, 1999.
- [6] R. Greenfeld. Translational tilings: structured or wild? In *Proceedings of the International Congress of Mathematicians 2026—Volume 3: Invited Lectures (Sections 1–4)*, pages 76–96. SIAM, 2026.
- [7] R. Greenfeld and T. Tao. The structure of translational tilings in $\mathbb{Z}^d$. *Discrete Anal.*, pages Paper No. 16, 28, 2021.
- [8] R. Greenfeld and T. Tao. Undecidable translational tilings with only two tiles, or one nonabelian tile. *Discrete Comput. Geom.*, 70(4):1652–1706, 2023.
- [9] R. Greenfeld and T. Tao. A counterexample to the periodic tiling conjecture. *Ann. of Math. (2)*, 200(1):301–363, 2024.
- [10] R. Greenfeld and T. Tao. Undecidability of translational monotilings. *J. Eur. Math. Soc.*, 2025. Published online June 21, 2025.
- [11] E. Jeandel and M. Rao. An aperiodic set of 11 Wang tiles. *Adv. Comb.*, pages Paper No. 1, 37, 2021.
- [12] E. Jeandel and N. Rolin. Fixed parameter undecidability for Wang tilesets. *Electron. Proc. Theor. Comput. Sci.*, 90:69–85, 2012.
- [13] E. Jeandel and P. Vanier. The undecidability of the domino problem. In *Substitution and Tiling Dynamics: Introduction to Self-Inducing Structures*, volume 2273 of *Lecture Notes in Math.*, pages 293–357. Springer, Cham, 2020.
- [14] A. S. Kahr, E. F. Moore, and H. Wang. Entscheidungsproblem reduced to the $\forall\exists\forall$ case. *Proc. Natl. Acad. Sci. USA*, 48(3):365–377, 1962.
- [15] Y. Kim. Undecidability of translational tiling with 2 polycubes, 2025. Preprint, [arXiv:2508.11725](https://arxiv.org/abs/2508.11725).
- [16] J. C. Lagarias and Y. Wang. Tiling the line with translates of one tile. *Invent. Math.*, 124(1–3):341–365, 1996.
- [17] T. Meyerovitch, S. Sanadhya, and Y. Solomon. A note on reduction of tiling problems. *Israel J. Math.*, 267(1):421–435, 2025.
- [18] D. J. Newman. Tessellation of integers. *J. Number Theory*, 9(1):107–111, 1977.
- [19] N. Ollinger. Tiling the plane with a fixed number of polyominoes. In *Language and Automata Theory and Applications*, volume 5457 of *Lecture Notes in Comput. Sci.*, pages 638–647, Berlin, 2009. Springer.
- [20] R. M. Robinson. Undecidability and nonperiodicity for tilings of the plane. *Invent. Math.*, 12(3):177–209, 1971.
- [21] J. B. Rosser. Extensions of some theorems of Gödel and Church. *J. Symbolic Logic*, 1(3):87–91, 1936.
- [22] M. Szegedy. Algorithms to tile the infinite grid with finite clusters. In *Proceedings of the 39th Annual Symposium on Foundations of Computer Science*, pages 137–145, Los Alamitos, CA, 1998. IEEE Computer Society.
- [23] H. Wang. Proving theorems by pattern recognition—II. *Bell System Tech. J.*, 40(1):1–41, 1961.
- [24] C. Yang and Z. Zhang. Undecidability of translational tiling of the plane with four tiles, 2025. Preprint, [arXiv:2506.19295](https://arxiv.org/abs/2506.19295).